MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

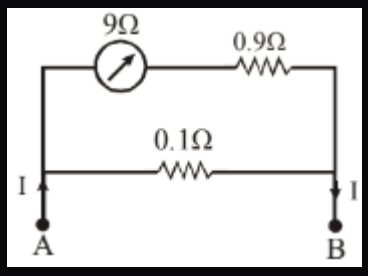
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A milliammeter of range $10\ \text{mA}$ and resistance 9Ω is joined in a circuit as shown.
The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B. The value of I is :-



(A) $100\ \text{mA}$

(B) $900\ \text{mA}$

(C) $1\ \text{A}$

(D) $1.1\ \text{A}$

Q.2

The de Broglie wavelength of an electron moving with a velocity $1.5 \times 10^8\ \text{ms}^{-1}$ is equal to that of a photon. The ratio of the kinetic energy of the electron to that of the energy of photon is :

(A) 1.5

(B) 1

(C) 0.5

(D) 0.25

Q.3

Which of the following is the unit of mobility of a electron in a conductor?

(A) $\mathrm{kg}^{-1}\ \mathrm{s}^{2}\ \mathrm{A}^{-1}$

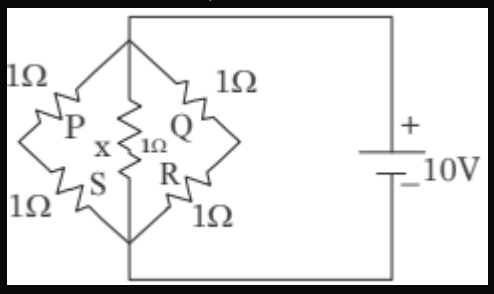
(B) $\mathrm{kg}^{-1}\ \mathrm{s}^{2}\ \mathrm{A}$

(C) $\mathrm{kg}^{-1}\ \mathrm{ms}^{2}\ \mathrm{A}^{-1}$

(D) $\mathrm{kgms}^{-1}\ \mathrm{A}^{-1}$

Q.4

Choose the correct statement in the shown circuit: (all the resistors are of $1\mathrm{ohms}$ x: is the Resistor in the middle also of $1\mathrm{ohms}$)



(A) The maximum heat loss is observed in resistor " $\mathbf{x}$ ".

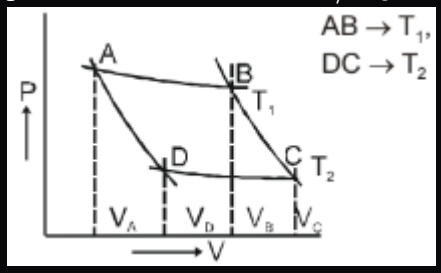
(B) The current through Battery is $10\ \mathrm{A}$.

(C) The current through resistor " $\mathbf{x}$ " is zero.

(D) None of these

Q.5

In the following $\mathrm{P}-\mathrm{V}$ diagram of an ideal gas, AB and CD are isothermal where as BC and DA are adiabatic process. The value of V_B/V_C is



(A) $=\mathrm{V_A}/\mathrm{V_D}$

(B) $<\mathrm{V_A}/\mathrm{V_D}$

(C) $>\mathrm{V_A}/\mathrm{V_D}$

(D) cannot say

Q.6

A Zener diode voltage regulator operated in the range $120\ \mathrm{V}-180\ \mathrm{V}$ produces a constant supply of $110\ \mathrm{V}$ and $250\ \mathrm{mA}$ to the load. If the maximum current is equally shared between the load and the Zener diode, then the values of load resistance (R_L) and series resistance (R_S) are respectively

(A) $R_L=280\Omega, R_S=70\Omega$

(B) $R_L = 440\Omega, R_S = 140\Omega$

(C) $R_L = 70\Omega, R_S = 280\Omega$

(D) $R_L = 440\Omega, R_S = 1400\Omega$

Q.7

In vernier callipers instrument 20 vernier scale divisions concide with 18 main scale divisions where $1\ \text{mm} = 1$ main scale division. The least count is

(A) $0.02\ \text{mm}$

(B) $0.05\ \text{mm}$

(C) $0.1\ \text{mm}$

(D) $0.2\ \text{mm}$

Q.8

Relative density of the material of a body is found by weighing it first in air and then in water. The weight in air is (5.00 ± 0.05) newton and weight in water is (4.00 ± 0.05) newton. Then the relative density of material should be:-

(A) $5.0 \pm 11\%$

(B) $5.0 \pm 1\%$

(C) $5.0 \pm 6\%$

(D) $1.25 \pm 5\%$

Q.9

Young's modulus for perfectly rigid body is

(A) zero

(B) infinite

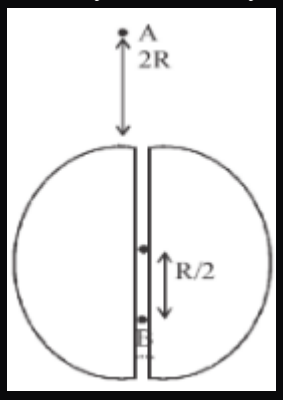
(C) $1 \times 10^{10}\text{Nm}^{-2}$

(D) $10 \times 10^{10}\text{Nm}^{-2}$

Q.10

Suppose, if a tunnel is dug along the diameter of the earth and a body of mass m is released from a point ' A ' at a distance $2R$ above earth along the line of tunnel. If M is the mass of the earth & R is the radius of the earth. The

velocity of the body during its fall when it crosses the point B at a distance $\frac{R}{2}$ below the earth centre as shown.



(A) $\frac{5}{2}\sqrt{\frac{GM}{3R}}$

(B) $\sqrt{\frac{GM}{3R}}$

(C) $2\sqrt{\frac{GM}{3R}}$

(D) $\frac{3}{2}\sqrt{\frac{GM}{3R}}$

Q.11

Energy levels A, B and C of a certain atom correspond to increasing values of energy, i.e. $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are the wavelengths of radiations corresponding to transitions C to B, B to A and C to A respectively, which of the following relations is correct?

(A) $\lambda_3 = \lambda_1 + \lambda_2$

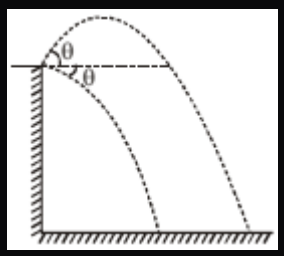
(B) $\lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

(C) $\lambda_1 + \lambda_2 + \lambda_3 = 0$

(D) ${\lambda_3}^2 = {\lambda_1}^2 + {\lambda_2}^2$

Q.12

Two balls are thrown with the same speed v_0 from the top of a cliff. The angles of their initial velocities are θ above and below the horizontal, as shown. How much farther along the ground does the top ball hit than the bottom ball :-



(A) $\frac{2v_0^2 \sin^2\theta\cos^2\theta}{g}$

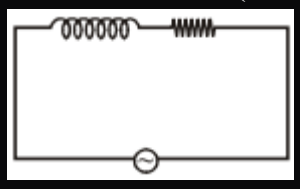
(B) $\frac{2v_0^2 \sin\theta}{g}$

(C) $\frac{2v_0^2 \cos\theta}{g}$

(D) $\frac{2v_0^2 \sin\theta\cos\theta}{g}$

Q.13

In a series LR circuit connected to an alternating voltage source, it is observed that at the instant voltage across the source is maximum, voltage across the inductor is $3\ \text{V}$ and voltage across the resistance is $4\ \text{V}$. If resistance is 2Ω, what is reactance (in Ω) of the inductor ?



(A) $\sqrt{3}$

(B) $\sqrt{2}$

(C) $2\sqrt{2}$

(D) $2\sqrt{3}$

Q.14

A ball, moving with a speed of $9\ \text{m/s}$, strikes an identical stationary ball such that after the collision, the direction of each ball makes an angle of 30° with the original line of motion. The speeds of the two balls after the collision are, respectively,

(A) $3\sqrt{3}\ \text{m/s}, 3\ \text{m/s}$

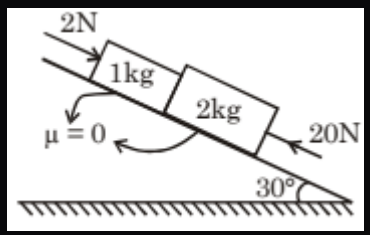
(B) $3\ \text{m/s}, 3\sqrt{3}\ \text{m/s}$

(C) $3\ \text{m/s}, 3\ \text{m/s}$

(D) $3\sqrt{3}\ \text{m/s}, 3\sqrt{3}\ \text{m/s}$

Q.15

The normal reaction force between the blocks and its direction will be :-



(A) $8\ \text{N}$, up the incline

(B) $8\ \text{N}$, down the incline

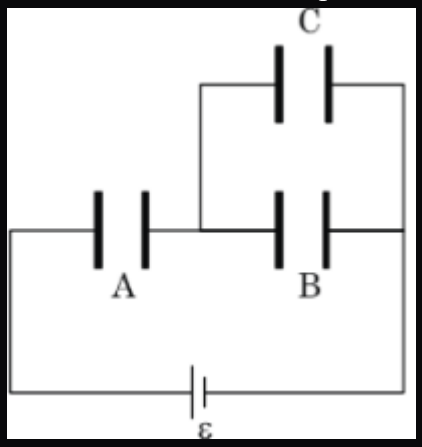
(C) $5\ \text{N}$, up the incline

(D) $5\ \text{N}$, down the incline

Q.16

Three capacitors A, B and C are connected with battery of emf ε. All capacitors are identical initially. If dielectric

slab is inserted between plates of capacitor A slowly with help of external force then



(A) Chemical energy of battery remains unchanged

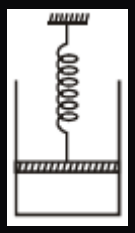
(B) Positive work is done on the given system by external force.

(C) Force between plates of capacitor C decreases.

(D) Magnitude of change of charge on B and C is same.

Q.17

An ideal monoatomic gas is enclosed in a vertical cylinder of cross section area A and length ℓ, under frictionless piston connected to spring of spring constant K. Atmosphere pressure is such that in absence of gas under the piston, equilibrium of piston is achieved when piston just touches the bottom of the cylinder and spring is in stretched configuation. Q amount of heat is supplied to gas slowly to move piston to the upper edge. Initially volume occupied by gas is $\frac{1}{3}\mathrm{rd}$ of the volume of cylinder. If $\mathrm{Q}=2\lambda\frac{\mathrm{K}\ell^2}{9}$, find λ, (through the motion of the piston spring is in stretched configuration)



(A) 6

(B) 4

(C) 8

(D) 2

Q.18

In an LRC series circuit at resonance current in the circuit is $10\sqrt{2}\ \mathrm{A}$. If now frequency of the source is changed such that now current lags by 45° than applied voltage in the circuit, which of the following is correct :

(A) Frequency must be increased and current after the change is $10\ \mathrm{A}$

(B) Frequency must be decreased and current after the change is $10\ \mathrm{A}$

(C) Frequency must be decreased and current is same as that of initial value

(D) The given information is insufficient to conclude anything

Q.19

The diameter of a flywheel is $1\ \mathrm{m}$. It has a mass of $20\ \mathrm{kg}$. It is rotating about its axis with a speed of 120 rotations in one minute. Its angular momentum in $\mathrm{kg-m^2/s}$ is

(A) 13.4

(B) 31.4

(C) 41.4

(D) 43.4

Q.20

Assertion A : For light diverging from a point source, the intensity at the wavefront does not depend on the distance.
Reason R : In a diverging beam of light from a point source, spherical wavefront is observed.

(A) (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true.

Q.21

Two sound waves of slightly different frequencies have amplitude ratio $\frac{11}{9}$. What is the difference of sound levels in decibels of maximum and minimum intensities heard at a point?

Q.22

Two bodies of mass $4\ \mathrm{kg}$ and $5\ \mathrm{kg}$ are moving along east and north directions with velocities $5\ \mathrm{m/s}$ and $3\ \mathrm{m/s}$ respectively. Magnitude of the velocity of centre of mass of the system is Vcm, find 18 Vcm.

Q.23

A spherical soap bubble has internal pressure P_0 and radius $\mathrm{r_0}$ and is in equilibrium in an enclosure with pressure $\mathrm{P_1 = \frac{8P_0}{9}}$. The enclosure is gradually evacuated. Assuming temperature and surface tension of soap bubble to befixed find the value of $\frac{\text{final radius}}{\text{initial radius}}$ of soap bubble.

Q.24

What is the force/length between two long parallel current carrying wires carrying current of $186\ \mathrm{A}$ and $24\ \mathrm{A}$ at a distance of $0.1\ \mathrm{mm}$ from each other. Mark answer to nearest integer (in $\mathrm{N/m}$).

Q.25

The radius of germanium (Ge) nuclide is measured to be twice the radius of $^{9}_{4}Be$. The number of nucleons in Ge will be

Q.26

Which of the following is not correctly matched for enzymatic reactions?

(A) Proteins $\rightarrow$ Amino acids ; Trypsin

(B) Starch $\rightarrow$ Maltose ; Diastase

(C) Sucrose $\rightarrow$ glucose and fructose; zymase

(D) Maltose $\rightarrow$ glucose; Maltase

Q.27

Assertion (A) : 1- Bromopentane reacts with $AgCN$ to give pentylisocyanide
Reason (R): $AgCN$ is mainly ionic in nature

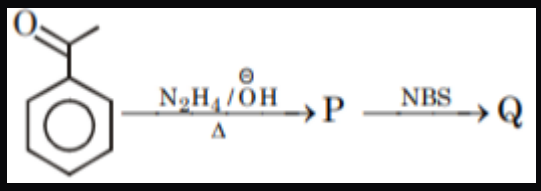
(A) A is true, R is true and R is correct explanation of A

(B) A is true, R is true but R is not correct explanation of A

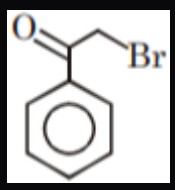
(C) A is true, R is false

(D) A is false, R is true

Q.28

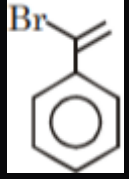


is :

(A) 

(B) 

(C) 

(D) 

Q.29

Which one of the following has maximum number of hybrid orbitals?

(A) C_6H_6

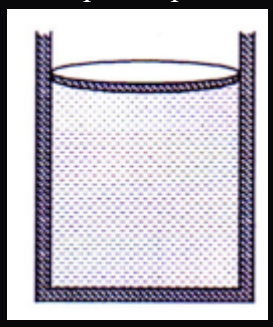
(B) $(CH_3)_4\,C$

(C) $(CH_3)_2\,C=O$

(D) $CH_3-CH=CH-CN$

Q.30

An adiabatic cylinder has 8 gram of helium in it at $400\ \mathrm{K}$ and its piston of area $1\ \mathrm{cm}^2$ (which has mass $1\ \mathrm{kg}$) is movable. If a block of mass $62\ \mathrm{kg}$ is placed on the piston then the final temperature of the gas will be (Take atmospheric pressure to be $10^5\mathrm{Nm}^{-2}$ and $g=10\ \mathrm{m/s}^2$)



(A) 500 K

(B) 800 K

(C) 1200 K

(D) 1600 K

Q.31

Polarization means "the distortion of the shape of an anion by an adjacently placed cation". Which of the following statements about polarization is correct

(A) Maximum polarization is brought about by a cation of high charge

(B) Minimum polarization is brought about by a cation of low radius

(C) A large cation is likely to bring about a large degree of polarization

(D) A small anion is likely to undergo a large degree of polarization

Q.32

Using Duma's method for estimating nitrogen $1\,g$ of an organic compound X gave $82\,mL$ of nitrogen, collected at $27^{\circ}C$ and $750\,mmHg$ pressure. If the aqueous tension at $27^{\circ}C$ is $30\,mmHg$ pressure, then the percentage of nitrogen in the given compound X is

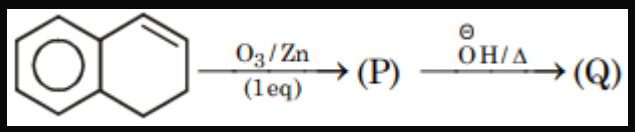
(A) 22.09 %

(B) 88.36 %

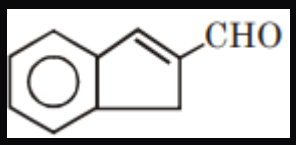
(C) 44.18 %

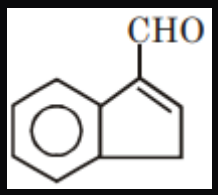
(D) 70.69 %

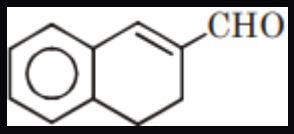
Q.33

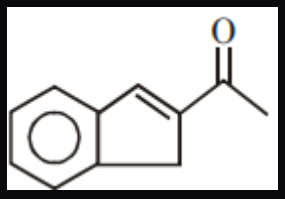


is :

(A) 

(B) 

(C) 

(D) 

Q.34

Match the following.

List I	List II
A. The reaction of 1, 6-dibromo hexane with Zn.	i. $H_3C—C \equiv CH$
B. Reaction of ethanol with concentrated H_2SO_4 at 443 K.	ii. $H_2C=CH_2$
C. Major product in the reaction of propene with HBr in the presence of benzoyl peroxide.	iii. 
D. The reaction of 1, 1-dibromopropane with $NaNH_2$ at 433 K.	iv. 

The correct answer is

	A	B	C	D
(a)	(iv)	(ii)	(iii)	(i)
(c)	(ii)	(iii)	(i)	(iv)

	A	B	C	D
(b)	(iii)	(i)	(ii)	(iv)
(d)	(i)	(ii)	(iv)	(iii)

(A) (a)

(B) (b)

(C) (c)

(D) (d)

Q.35

Which one of following is a disproportionation reaction?

(A) $2AgNO_3(aq) + Cu(s) \longrightarrow Cu\,(NO_3)_2\,(aq) + 2Ag(s)$

(B) $3AgNO_3(aq) + K_3PO_4(aq) \longrightarrow Ag_3PO_4(\,s) + 3KNO_3(aq)$

(C) $4KClO_3(\,s) \xrightarrow{\Delta} KCl(s) + 3KClO_4(\,s)$

(D) $4Fe(s) + 3O_2(\,g) \longrightarrow 2Fe_2O_3$

Q.36

Observe the following reaction

$Cl_2(\,g) + 2OH^-(aq) \rightarrow ClO^-(aq) + Cl^-(aq) + H_2O(l)$

Identify the correct statements about this reaction

I. $Cl_2(\,g)$ is oxidized to ClO^-(aq)

II. $Cl_2(\,g)$ is oxidized to $Cl^-(aq)$

III. $Cl_2(\,g)$ is reduced to ClO^-(aq)

IV. $Cl_2(g)$ is reduced to $Cl^-(aq)$

The correct answer is

(A) I, IV only

(B) I, III only

(C) I, II, IV only

(D) II, III only

Q.37

The set with only ambidentate ligands in the following is

(A) $NO_3^{\ominus}, Br^{\ominus}, C_2O_4^{2\ominus}$

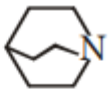
(B) $NO_2^{\ominus}, CN^{\ominus}, SCN^{\ominus}$

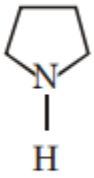
(C) $NO_2^{\ominus}, C_2O_4^{2\ominus}, NH_3$

(D) $SCN^{\ominus}, CO, NH_3$

Q.38

More basic between following pairs of compounds?

(a) (I) $(C_2H_5)_3N$ and (II) 

(b) (I) $CH_3CH_2NHCH_2CH_3$ & (II) 

(c) (I) $H_2N-\overset{\overset{O}{\|}}{C}-NH_2$ & (II) $NH_2-\overset{\overset{NH}{\|}}{C}-NH_2$

(A) (a) - (I), (b) - (II), (c) - (I)

(B) (a) - (I), (b) - (I), (c) - (I)

(C) (a) - (II), (b) - (I), (c) - (II)

(D) (a) - (II), (b) - (II), (c) - (II)

Q.39

How many of the following statements are correct?

(a) 'He' is the second most abundant element in the universe.

(b) the symbol for the element with atomic number 110 is Ds.

(c) Osmium has the highest density among all elements.

(d) Francium is the most electropositive element in the periodic table.

(A) 3

(B) 2

(C) 4

(D) 1

Q.40

A sample of hydrogen atoms de-excites from 6 th excited state to the ground state in one or more electronic transitions. If no spectral line is obtained in the Paschen and Brackett series, then the maximum number of spectral lines of different photon energies obtained will be -

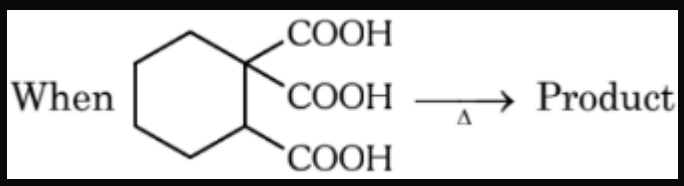
(A) 21

(B) 15

(C) 10

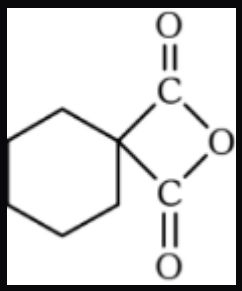
(D) 7

Q.41

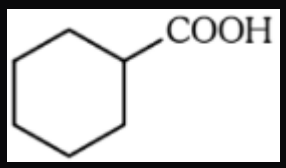


The product formed is

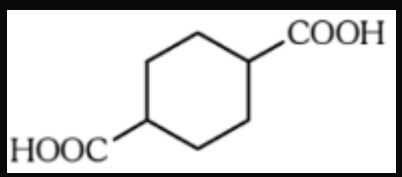
(A)



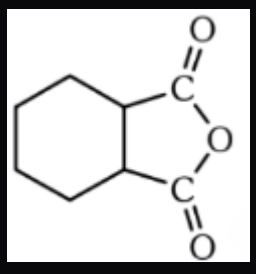
(B)



(C)



(D)



Q.42

Cold ferrous sulphate solution on absorptive of NO develops brown colour due to the formation of

(A) paramagnetic $[Fe(H_2O)_5(NO)]SO_4$

(B) diamagnetic $[Fe(H_2O)_5(N_3)]SO_4$

(C) paramagnetic $[Fe(H_2O)_5(NO_3)][SO_4I_2$

(D) diamagnetic $[Fe(H_2O)_4(SO_4)]NO_3$

Q.43

In an experiment $20cc$ of a semimolar H_2SO_4 solution was added to $30cc$, of a seminormal $AgNO_3$ solution. Ag_2SO_4 was precipitated out and the excess acid was titrated against a normal (1 N) $NaOH$ solution. The volume of $NaOH$ required is

(A) $5\ mL$

(B) $10\ mL$

(C) $20\ mL$

(D) $30\ mL$

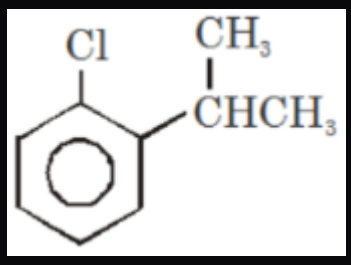
Q.44

$$C_2H_2 \xrightarrow[\text{Iron tube}]{\text{Red Hot}} A \xrightarrow{Cl_2,\ \text{Anhy. } AlCl_3} B$$

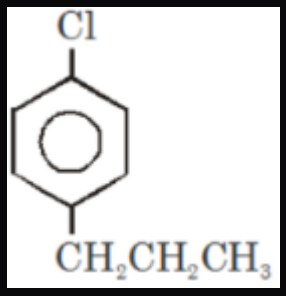
$$B \xrightarrow[(2)\ Zn(Hg)/HCl]{(1)\ CH_3CH_2COCl,\ \text{Anhy. } AlCl_3} C$$

The major product ' C ' is

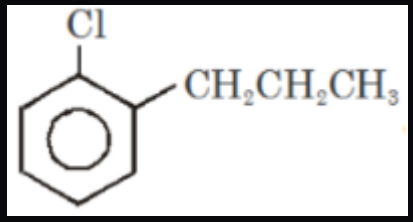
(A)



(B)



(C)



(D)



Q.45

In a group separation operation, before precipitating out group III metal ions as hydroxides, it is necessary to boil the solution of the salt mixture with a few milliliters of conc. HNO_3. This is done to convert

(A) Co^{2+} to Co^{3+}

(B) Fe^{2+} to Fe^{3+}

(C) Mn^{2+} to MnO_4^*

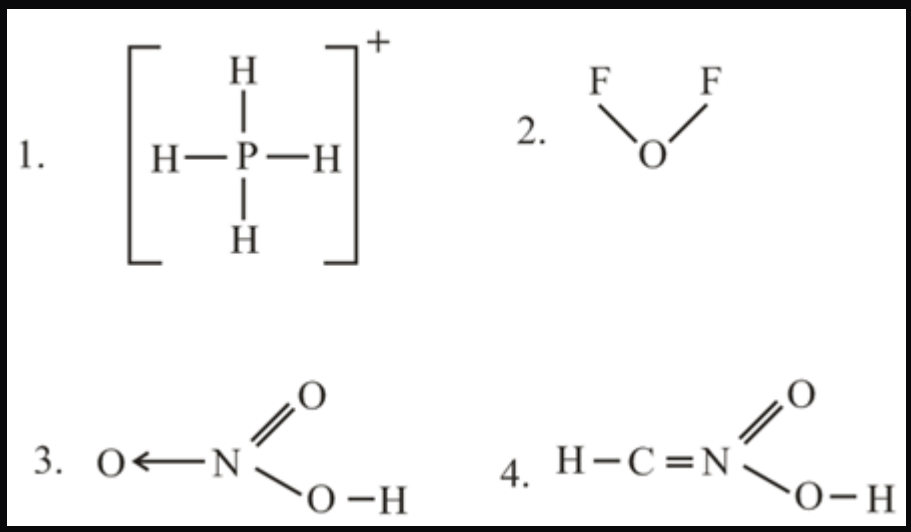
(D) Cr^{3+} to CrO_4^{2-}

Q.46

An organic compound $\mathbf{P}$ contains 62.07% carbon and 10.34% hydrogen and rest oxygen. Its vapour density is 29 . This compound does not react with sodium metal, but its $2.9\ g$ combines with Xg of bromine (to give dibromo addition product). Find out value of $(Y - X)$. (Where Y is total number of possible isomers of given organic compound P)
[Atomic mass : $Br = 80$]

Q.47

In how many representation of the below-mentioned compounds, does not correctly represent the bonding capacities of the atoms involved?



Q.48

How many of the following compounds are associated with $sp^3 - s - sp^3$ type of overlap?
(A) $Al_2(CH_3)_6$

(B) Be_2H_4
(C) $(BeH_2)_n$
(D) Al_2H_6

Q.49

For a 1st order reaction (gaseous) (cont. V, T) $aA \longrightarrow (b-1)B + C$ (with $b > a$) the pressure of the system increases by $50\left(\frac{b}{a}-1\right)\%$ in a time of $10\ min$. The half life of the reaction is therefore _____ min.

Q.50

$5\ mL$ of a gaseous hydrocarbon was exposed to $30\ mL$ of O_2. The resultant gas, on cooling is found to measure $25\ mL$, of which $10\ mL$ was absorbed by $NaOH$ and the remainder by pyrogallol. All measurements are made at constant pressure and temperature. The sum of number of carbon and hydrogen atom in the hydrocarbon is

Q.51

If $f(x)$ is a non constant polynomial function $f : R \to R$ such that $7\frac{d}{dx}(xf(x)) = 3f(x) + 4f(x+1)$, $f(-1) + f(0) = 2$, then number of such functions is

(A) 0

(B) 1

(C) 2

(D) 3

Q.52

If point $P(\cos\theta, \cos\theta)$ lies outside the circle $x^2 + y^2 + 2x + y - 2 = 0$, then the total number of values of n which exist if $\theta = \frac{n\pi}{6}, n \in N$ and $\theta \in [0, 2\pi]$, are -

(A) 0

(B) 3

(C) 5

(D) 7

Q.53

The complete interval in which the function $f(x) = \frac{2^x + 2^{1-x} - 1}{2^{-x} + 1}, x \in R$ increases, is

(A) $(0, \infty)$

(B) $(-\infty, 0)$

(C) $\left(-\infty, \frac{1}{2}\right)$

(D) $(1, \infty)$

Q.54

Solution of differential equation $\frac{dx}{dy} - \frac{x \log x}{1+\log x} = \frac{e^y}{1+\log x}$, if $y(1) = 0$, is

(A) $\mathrm{e^y = x^{e^y}}$

(B) $x = ye^y$

(C) $x^x = e^{ye^y}$

(D) $y = e^{x \cdot e^y}$

Q.55

Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose det. $A = 6$. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ then

(A) det. $B = 6$

(B) det. $B = -6$

(C) det. $B = 12$

(D) det. $B = -12$

Q.56

If $a, b, c, x, y, z \in R$ such that $a^2 + b^2 + c^2 = 4$ and $x^2 + y^2 + z^2 = 9$ then find maximum value of $(az - cx)^2 + (ay - bx)^2 + (bz - cy)^2$

(A) 18

(B) 6

(C) 36

(D) None of these

Q.57

Consider, $f(x) = \frac{(x^2-1)(x^2-4)}{g(x)}, x \in R - \{a, b\}$, where $\mathrm{g(x)}$ is a polynomial of degree ≤ 4.
$\mathrm{x = a, b}$ are not in the domain of $f(\mathrm{x})$ but limit at these point exist.
The following three informations are also known
(i) $\mathrm{Lim}_{x \to \pm\infty} f(x) = +\infty$
(ii) $\mathrm{Lim}_{x \to -1} f(x) = 6$
(iii) $\mathrm{Lim}_{x \to -2} f(x) =$ a non-zero finite number
Let L and M denotes the number of points of discontinuity and the number of points of non-differentiability of $y = |f(x)|$ respectively in R (the set of all real numbers). the value of $(\mathrm{L} + \mathrm{M})$ is

(A) 4

(B) 2

(C) 3

(D) 5

Q.58

Let a, b, c, d be four distinct real numbers in A.P. Find the smallest positive value of k satisfying
$2(a-b)+k(b-c)^2+(c-a)^3=2(a-d)+(b-d)^2+(c-d)^3$

(A) 18

(B) 20

(C) 16

(D) 15

Q.59

Let $x_1, x_2, x_3 \ldots x_k$ be k observations and $w_i = ax_i + b$ for $i = 1, 2, 3 \ldots \ldots k$, where a and b are constants. If mean of x_i is 52 and their standard deviation is 12 and mean of w_i is 60 and their standard deviation is 15 , then the value of a and b should be

(A) $a = 1.25,\ b = -5$

(B) $a = -1.25,\ b = 5$

(C) $a = 2.5,\ b = -5$

(D) $a = 2.5,\ b = 5$

Q.60

Area bounded by $f(\mathrm{x}) = \min(\{\mathrm{x}-2\}, \{3-\mathrm{x}\}), \mathrm{x} = 4, \mathrm{x} = -4$ and x-axis is equal to (where $\{.$ denotes fractional part)

(A) 2.5

(B) 2

(C) 1.5

(D) 1

Q.61

Let $f(x) = x^2 + ax + b$. If $\forall x \in R$, there exist a real value of y such that $f(y) = f(x) + y$, then find the maximum value of $100a$

(A) 50

(B) 200

(C) 120

(D) 100

Q.62

The shortest distance between the lines $\bar{r} = (1-t)\hat{\imath} + (t-2)\hat{\jmath} + (3-2t)\hat{k}$ and $\bar{r} = (p+1)\hat{\imath} + (2p-1)\hat{\jmath} + (2p+1)\hat{k}$ is

(A) $\frac{8}{\sqrt{29}}$ units

(B) $\frac{4}{\sqrt{29}}$ units

(C) $\frac{2}{\sqrt{5}}$ units

(D) $\frac{4}{\sqrt{19}}$ units

Q.63

The mirror image of the curve $\arg\left(\frac{z-3}{z-i}\right) = \frac{\pi}{6}$ in the real axis is

(A) $\arg\left(\frac{z+3}{z+i}\right) = \frac{\pi}{6}$

(B) $\arg\left(\frac{z-3}{z+i}\right) = \frac{\pi}{6}$

(C) $\arg\left(\frac{z+i}{z+3}\right) = \frac{\pi}{6}$

(D) $\arg\left(\frac{z+i}{z-3}\right) = \frac{\pi}{6}$

Q.64

An ellipse with major axis and minor axis of length $10\sqrt{3}$ and 10 respectively, slides along the coordinate axes and always remains confined in the first quadrant. The locus of centre of the ellipse will be the arc of a circle.
The length of this arc will be equal to

(A) 10π

(B) 5π

(C) $\frac{5\pi}{4}$

(D) $\frac{5\pi}{3}$

Q.65

If $\mathrm{A}^2 = \mathrm{A}, \mathrm{B}^2 = \mathrm{B}$ and $\mathrm{AB} = \mathrm{BA}$, then the number of distinct real values that $\mathrm{Det}(\mathrm{A} - \mathrm{B})$ can take-

(A) 2

(B) 3

(C) 4

(D) Infinitely many

Q.66

Find the number of all 6 digit numbers such that
(a) All the digits of each number are from the set $\{1, 2, 3, 4, 5\}$
(b) Any digit that appears in the number appeares at least twice.

(A) 1405

(B) 1512

(C) 1205

(D) 1312

Q.67

A person has to catch a train. To catch train, from his home he can take a taxi or take rickshaw or walk by foot with respective probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$. Probability of him catching train when he takes rickshaw from his home is half that of when the takes the taxi and probability of catching the train when he walked by foot is $\frac{1}{4}$ th that of when he takes rickshaw. He finally reached the train, the probability he walked by his foot to catch the train, is

(A) $\frac{1}{33}$

(B) $\frac{2}{33}$

(C) $\frac{7}{33}$

(D) $\frac{13}{33}$

Q.68

The number of points (x, y) satisfying $x^2 + y^2 \leq 4 \ \& \ \tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ is equal to

(A) 4

(B) 2

(C) 3

(D) 5

Q.69

For each positive integer n, consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$. Then, $\lim_{n \to \infty} (n \cdot d_n)$ has the value equal to

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) 0

Q.70

$\int_{-1}^{1} \{x^2 + x - 3\} \, dx$, where $\{x\}$ denotes the fractional part of x, is equal to

(A) $\frac{1}{3}(1 + 3\sqrt{5})$

(B) $\frac{1}{6}(1 + 3\sqrt{5})$

(C) $\frac{1}{3}(3\sqrt{5} - 1)$

(D) $\frac{1}{6}(3\sqrt{5}-1)$

Q.71

If $x\,dy = y(\,dx + y\,dy), y(1) = 1, y(x) > 0$, then $y(-3)$ is

Q.72

It is given that complex numbers z_1 and z_2 satisfy $|z_1| = 2$ and $|z_2| = 3$. If the included angle of their corresponding vectors is 60° then $\left|\frac{z_1+z_2}{z_1-z_2}\right|$ can be expressed as $\frac{\sqrt{N}}{7}$ where N is natural number then N equals

Q.73

$\mathrm{P(x)}$ is a polynomial such that $P(x) + P(2x) = 5x^2 - 18$, then $\lim_{x\to 3}\frac{P(x)}{x-3}$

Q.74

If Q is the foot of the perpendicular from the point $P(4,-5,3)$ on the line $\frac{x-5}{3} = \frac{y+2}{-4} = \frac{z-6}{5}$ then $25(PQ)^2 - 450 =$

Q.75

Let $S = \{1, 2, 3, \ldots 9\}$. For $k = 1, 2, \ldots 5$, let N_k be the number of subsets of S, each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$

Answers & Solutions

Q.1 Answer:

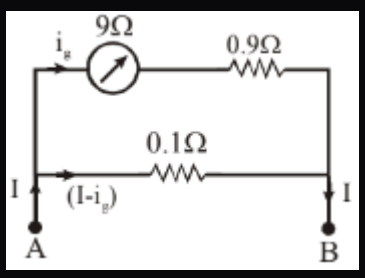
1 A

Solution:

$\mathrm{i_g} = 10\ \mathrm{mA} = 0.01\ \mathrm{A}$

$\mathrm{V_A} - \mathrm{V_B} = (\mathrm{I} - \mathrm{i_g})\,0.1 = \mathrm{i_g} \times 9.9$

or $\mathrm{I} \times 0.1 = 10\mathrm{i_g}$



or $\mathrm{I} = \frac{10\times 0.01}{0.1} = 1\ \mathrm{A}.$

Q.2 Answer:

0.25

Solution:

$$\lambda_e = \frac{h}{m_e v_0} = \lambda \text{ photon } = \frac{hc}{E_{\text{photon}}}$$

$$\frac{\text{KEelectron}}{\text{Ephoton}} = \frac{h^2/2m\lambda^z}{hc/\lambda} = \frac{h}{2mc\lambda} = \frac{h}{2mc} \dot{h}^{mv} = \frac{v}{2c}$$

$$= \frac{1.5 \times 10^8}{6 \times 10^8} = 0.25$$

Q.3 Answer:

$\text{kg}^{-1}\ \text{s}^2\ \text{A}$

Solution:

Mobility is given as $\mu = \frac{e\tau}{\text{m}}$
So, unit of μ **is** $\frac{\text{coulomb} - \text{sec}}{\text{kg}}$ **i.e.** $\text{kg}^{-1}\ \text{s}^2\ \text{A}$

Q.4 Answer:

The maximum heat loss is observed in resistor " $\mathbf{x}$ ".

Solution:

$$\text{H}_\text{x} = \frac{(10)^2}{1}\text{t watt } = 100\text{t watt}$$

$$\text{H}_\text{P} = \text{H}_\text{Q} = \text{H}_\text{R} = \text{H}_\text{S} = \frac{(5)^2}{1}\text{t watt } = 25\text{t watt}$$

Q.5 Answer:

$= \text{V}_\text{A}/\text{V}_\text{D}$

Solution:

$$\text{AB} \rightarrow \text{ isothermal}$$
$$\text{P}_\text{A}\text{V}_\text{A} = \text{P}_\text{B}\text{V}_\text{B} \quad \ldots(i)$$
$$\text{BC} \rightarrow \text{ Adiabatic}$$
$$\text{P}_\text{B}\text{V}_\text{B}^\gamma = \text{P}_\text{C}\text{V}_\text{C}^\gamma \quad \ldots(ii)$$
$$\text{CD} \rightarrow \text{ Isothermal}$$
$$\text{P}_\text{C}\text{V}_\text{C} = \text{P}_\text{D}\text{V}_\text{D} \quad \ldots(iii)$$
$$\text{DA} \rightarrow \text{ Adiabatic}$$
$$\text{P}_\text{D}\text{V}_\text{D}^\gamma = \text{P}_\text{A}\text{V}_\text{A}^\gamma \quad \ldots(iv)$$

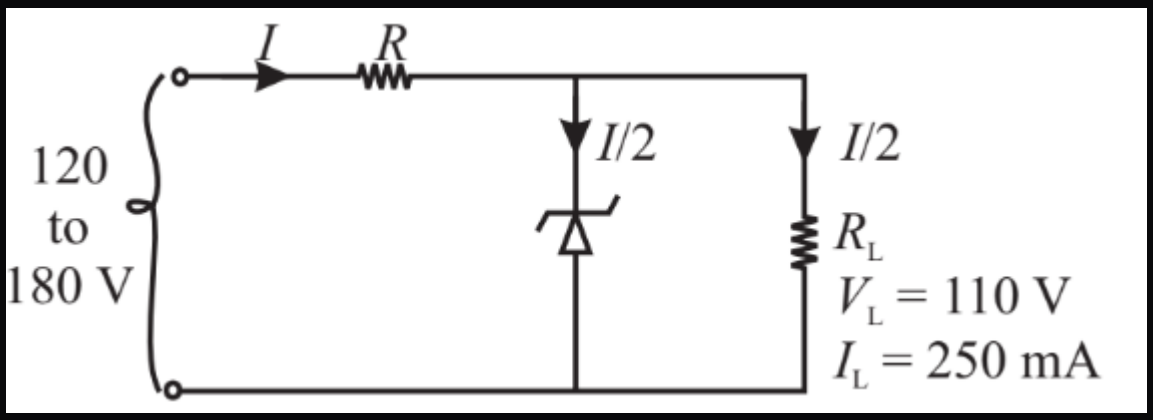
From (i), (ii), (iii) and (iv)

$\frac{V_B}{V_C} = \frac{V_A}{V_D}$

Q.6 Answer:

$R_L = 440\Omega, R_S = 140\Omega$

Solution:



We have, $R_L = \frac{V_L}{I_L} = \frac{110}{250 \times 10^{-3}} = 440\Omega$

As current is equally shared by Zener diode and resistance R. So, $I_Z = 250$ mA

Then, $I = 2 \times 250 = 500$ mA $\quad [\because I = I_L + I_Z] = 0.5$ mA

So, $\quad R_S = \frac{V_S}{I} = \frac{(180-110)}{0.5} = \frac{70}{0.5} = 140\Omega$

$[\because$ when current is maximum $V = 180$ V$]$

Q.7 Answer:

0.1 mm

Solution:

$$\text{L.C.} = \text{M.S.D} - \text{V.S.D}$$

$$20(VSD) = 18(MSD)$$

$$\Rightarrow VSD = \frac{18}{20}\text{MSD}$$

$$\therefore \text{ L.C.} = \left(1 - \frac{18}{20}\right)\text{MSD} = \frac{2}{20} \times 1 \text{ mm}$$

$$= 0.1 \text{ mm}$$

Q.8 Answer:

$5.0 \pm 11\%$

Solution:

$$F_{\mathrm{B}} = W - W'$$

$$= 1 \pm (0.1)$$

$$\frac{\rho_0 \mathrm{Vg}}{\rho_w \mathrm{Vg}} = \frac{5 \pm 0.05}{1 \pm 0.1}$$

$$\frac{\rho_0}{\rho_w} = \frac{5 \pm 0.05}{1 \pm 0.1}$$

$$= (5) \pm (\Delta\rho)$$

$$\frac{\Delta\rho}{5} = \frac{0.05}{5} + \frac{0.1}{1}$$

$$= 0.01 + 0.1 = 0.11$$

Q.9 Answer:

infinite

Solution:

In perfectly rigid body, there is no change in its dimensions on applying a deforming force, therefore strain in the perfectly rigid body is zero. $\therefore$ Young's modulus $= \frac{\text{Stress}}{\text{Strain}} = \frac{\text{Stress}}{0} = \infty$ Therefore, Young's modulus of a perfectly rigid body is infinite.

Q.10 Answer:

$\frac{5}{2}\sqrt{\frac{\mathrm{GM}}{3\mathrm{R}}}$

Solution:

$$V_A = \frac{-GM}{3R}$$

$$V_B = \frac{-GM}{2R^3}\left(3R^2 - \frac{R^2}{4}\right) = \frac{-11GM}{8R}$$

By conservation of energy

$$\mathrm{mV_A} = \mathrm{mV_B} + \frac{1}{2}\mathrm{mv}^2$$

$$\Rightarrow \quad \frac{-\mathrm{GMm}}{3\mathrm{R}} = \frac{-11\mathrm{GMm}}{8\mathrm{R}} + \frac{1}{2}\mathrm{mv}^2$$

$$\Rightarrow \quad v = \frac{5}{2}\sqrt{\frac{\mathrm{GM}}{3\mathrm{R}}}$$

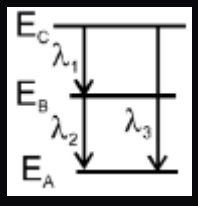
Q.11 Answer:

$\lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

Solution:

$$\mathrm{E_C} - \mathrm{E_A} = (\mathrm{E_C} - \mathrm{E_A}) + (\mathrm{E_B} - \mathrm{E_A})$$

$$\Rightarrow \frac{\mathrm{hc}}{\lambda_3} = \frac{\mathrm{hc}}{\lambda_1} + \frac{\mathrm{hc}}{\lambda_2}$$



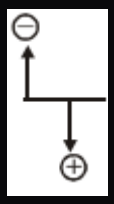
$\Rightarrow \lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}.$

Q.12 Answer:

$\frac{2v_0^2 \sin\theta\cos\theta}{g}$

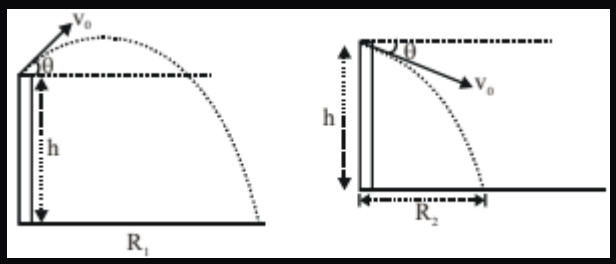
Solution:

For upper ball



$h = -v_0 \sin\theta t_1 + \frac{1}{2} g {t_1}^2$ **...(i)**

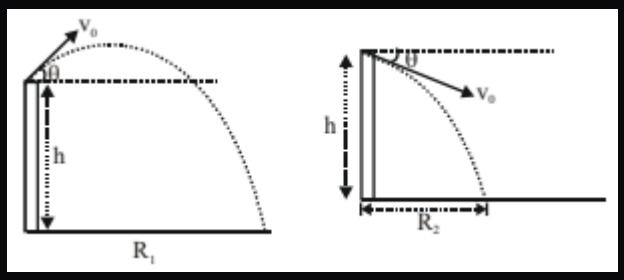
$R_1 = v_0 \cos\theta t_1$ **...(ii)**



For lower ball

$h = v_0 \sin\theta t_2 + \frac{1}{2} g {t_2}^2$ **...(iii)**

$R_2 = v_0 \cos\theta t_2$ **...(iv)**



From (i) - (iii)

$t_1 - t_2 = \frac{2v_0 \sin\theta}{g}$

From (ii) - (iv)

$R_1 - R_2 = \frac{2v_0^2 \sin\theta\cos\theta}{g}$

Q.13 Answer:

$\sqrt{3}$

Solution:

$V_L = i_0 X_L \sin(\omega t + 90 - \phi)$

$V_R = i_0 R \sin(\omega t - \phi)$

$V = V_0 \sin(\omega t)$

$3\,V = V_L = i_0 X_L \sin\phi$

$4\,V = V_R = i_0 R \cos\phi$

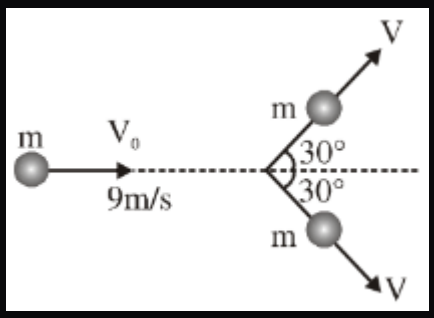
$\frac{3}{4} = \frac{\omega L}{R} \tan\phi = \frac{\omega^2 L^2}{R^2}$

$\Rightarrow \frac{\omega L}{R} = \frac{\sqrt{3}}{2} = \tan\phi$

Q.14 Answer:

$3\sqrt{3}\ \mathrm{m/s}, 3\sqrt{3}\ \mathrm{m/s}$

Solution:



Using momentum conservation

$$\mathrm{mv_0} = \mathrm{mvcos30^\circ} + \mathrm{mv\cos 30^\circ}$$

$$9 = \frac{\sqrt{3}}{2}\mathrm{v} \times 2 \Rightarrow 9 = \sqrt{3}\mathrm{v}$$

$$\Rightarrow \mathrm{v} = 3\sqrt{3}\ \mathrm{m/s}$$

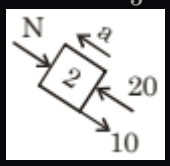
As there was no momentum along Y-axis the speed of each ball has to be same.

Q.15 Answer:

$8\ \mathrm{N}$, up the incline

Solution:

$$\mathrm{a} = \frac{20-2-15}{3} = 1\ \mathrm{m/s^2}$$



$$20 - 10 - N = 2 \times a$$

$$N = 10 - 2 = 8N$$

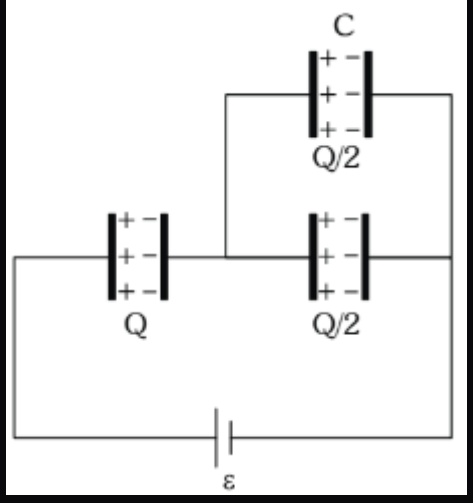
Q.16 Answer:

Magnitude of change of charge on B and C is same.

Solution:

Initial charge distribution is shown in figure.

$$\mathrm{Q} = \frac{2\mathrm{C}}{3}\varepsilon$$



Capacitance of A increases charge increases

Q.17 Answer:

8

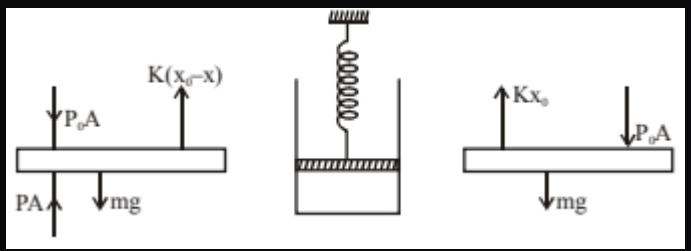
Solution:

$P_0\,A + mg = kx_0$

for very slow process

$PA = mg + p_0\,A - K(x_0 - x)$

$\Rightarrow P = \dfrac{Kx}{A} = \dfrac{KV}{A^2}$



Thus process followed by gas is $PV^{-1} = \frac{K}{A^2} =$ **constant.**

∴ Molar heat capacity for process is

$= C = C_v + \frac{R}{2} = 2R \quad \Rightarrow Q = n \cdot C\Delta T$

Since $\frac{P}{V} = \frac{AK}{A^2} \Rightarrow T = \frac{KV^2}{nRA^2}$

$\Rightarrow \Delta T = \frac{K}{nR}\left(1 - \frac{1}{9}\right)\ell^2 \Rightarrow Q = \frac{16\,K}{9}\ell^2$

Q.18 Answer:

Frequency must be increased and current after the change is $10\ A$

Solution:

Initially at resonance: $X_L = X_E \Rightarrow Z = R.$

$\therefore i_0 = \frac{\varepsilon_0}{R} = 10\sqrt{2}\ A$

After increasing frequency : $X_L > X_c$

$\omega L > 1/\omega C$

$\omega > \frac{1}{\sqrt{LC}} \Rightarrow \omega > \omega_0$ **(i.e frequency increases)**
and $i' = \frac{\varepsilon_0}{\sqrt{R^2+X^2}} = \frac{\varepsilon_0}{\sqrt{2}R} = i_0/\sqrt{2} = 10\text{amp.}$

Q.19 Answer:

31.4

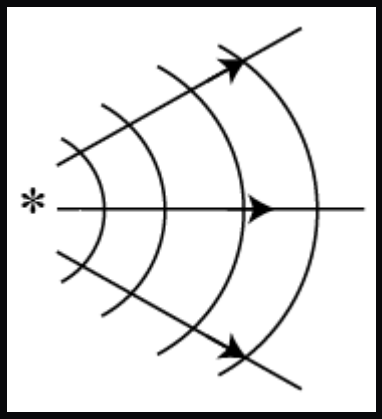
Solution:

$r = \frac{1}{2}\text{ m} = 0.5\text{ m}, m = 20\text{ kg},$
$n = 120\text{ rotation per minute},$
$= 2\text{ rotation per sec.}$
Angular momentum $L = I\omega$
$= \frac{Mr^2}{2}2\pi n = \frac{20 \times (0.5)^2}{2} \times 2 \times 3.14 \times 2$
$= 5 \times 3.14 \times 2 = 31.4\text{ kg} - \text{m}^2/\text{s}$

Q.20 Answer:

(A) is false but (R) is true.

Solution:



$I = \frac{P}{4\pi r^2} \propto \frac{1}{r^2}$

Q.21 Answer:

—

Solution:

$SL_1 - SL_2 = 10\log_{10}\left(\frac{I_{\max}}{I_{\min}}\right)$
$= 10\log_{10}\left(\frac{A_1 + A_2}{A_1 - A_2}\right)^2$
$\Rightarrow SL_1 - SL_2 = 20\log_{10}\left[\frac{\frac{A_1}{A_2} + 1}{\frac{A_1}{A_2} - 1}\right]$
$= 20\log_{10} 10 = 20\text{ dB}$

Q.22 Answer:

—

Solution:

Velocity of centre of mass is

$\mathbf{v}_{\mathrm{CM}}=\frac{m_1\mathbf{v}_1+m_2\mathbf{v}_2}{m_1+m_2}$

Given, $m_1=4\ \mathrm{kg}, m_2=5\ \mathrm{kg}, v_1=5\hat{\mathbf{j}}\mathrm{m/s}$

$v_2=3\hat{\mathbf{i}}\mathrm{m/s}$

$$\therefore \quad \mathbf{v}_{\mathrm{CM}}=\frac{4\times 5\hat{\mathbf{j}}+5\times 3\hat{\mathbf{i}}}{5+4}$$

$$=\frac{20\hat{\mathbf{i}}}{9}+\frac{15}{9}\hat{\mathbf{j}}$$

Hence, magnitude $|\mathbf{v}_{\mathrm{CM}}|=\sqrt{\left(\frac{20}{9}\right)^2+\left(\frac{15}{9}\right)^2}$

$=\frac{25}{9}\ \mathrm{m/s}$

Q.23 Answer:

—

Solution:

$\mathrm{P}_0-\frac{8\mathrm{P}_0}{9}=\frac{4\,\mathrm{S}}{\mathrm{r}_0}\Rightarrow \mathrm{P}_0=\frac{36\,\mathrm{S}}{\mathrm{r}_0}$

Finally $\mathrm{P}_{\text{outside}}=0\Rightarrow \mathrm{P}_{\text{inside}}=\frac{4\,\mathrm{S}}{\mathrm{r}}$

Now $\mathrm{P}_1\,\mathrm{V}_1=\mathrm{P}_2\,\mathrm{V}_2$

$\left(\frac{36\,\mathrm{S}}{\mathrm{r}_0}\right)\left(\frac{4}{3}\pi\mathrm{r}_0^3\right)=\left(\frac{4\,\mathrm{S}}{\mathrm{r}}\right)\left(\frac{4}{3}\pi\mathrm{r}^3\right)\Rightarrow \mathrm{r}=3\mathrm{r}_0$

Q.24 Answer:

—

Solution:

$$\mathrm{F}=\frac{\mu_0}{2\pi}\frac{\mathrm{i}_1\mathrm{i}_2}{\mathrm{r}}=2\times 10^{-7}\times\frac{186\times 24}{1\times 10^{-4}}$$

$=8.928\approx 9\ \mathrm{N/m}$

Q.25 Answer:

—

Solution:

Using radius of a nucleus, $\mathrm{R}\propto \mathrm{A}^{1/3}$

Where, $A=$ number of nucleons

$$\therefore \quad \frac{R_{Ge}}{R_{Be}} = \left(\frac{A_{Ge}}{A_{Be}}\right)^{1/3}$$

$$2 = \left(\frac{A_{Ge}}{A_{Be}}\right)^{1/3} \text{ or, } 2^3 = \frac{A_{Ge}}{A_{Be}} \quad (\therefore R_{Ge} = 2R_{Be})$$

$$2^3 = \frac{A_{Ge}}{9} \quad [\therefore A_{Be} = 9]$$

$$\therefore \quad A_{Ge} = 2^3 \times 9 = 8 \times 9 = 72$$

Q.26 Answer:

Sucrose $\rightarrow$ glucose and fructose; zymase

Solution:

Zymase $\Rightarrow$ enzyme for conversion of sugar to alcohol.
Invertase $\Rightarrow$ enzyme for conversion of sugar to its constituents glucose and fructose.

Q.27 Answer:

A is true, R is false

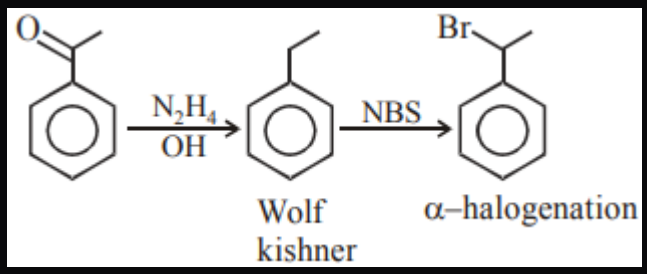
Solution:

AgCN is covalent in nature. Thats why haloalkane produces isocyanide compound with AgCN and cyanide compound with KCN.

Q.28 Answer:



Solution:



Q.29 Answer:

$(\mathrm{CH_3})_4\,\mathrm{C}$

Solution:

Hydrogen atom does not show hybridisation, thus number of hybrid orbitals in :
(a) $\mathrm{C_6H_6}$
All six C-atoms show sp^2-hybridisation (i.e. 3-orbitals by each C-atom)

$\therefore$ Total number of hybrid orbitals $= 6 \times 3 = 18$

(b) $(CH_3)_4\ C$ All five C-atoms show sp^3-hybridisation (i.e. 4-orbitals by each C-atom).

$\therefore$ Total number of hybrid orbitals $= 5 \times 4 = 20.$

(c) $(CH_3)_2\ C = O$

Two C-atoms belong to CH_3 group show sp^3-hybridisation (i.e. 4-orbital by each C-atom).

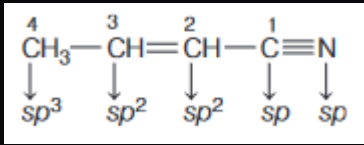
One C-atom, bonded with O-atom and $(CH_3)_2$-groups, show sp^2-hybridisation (i.e.3-hybrid orbitals).

One O- atom also show sp^2-hybridisation

Thus, total number of hybrid orbitals

$= 8 + 3 + 3 = 14$

(d)



- C-1 show sp-hybridisation (i.e. 2-hybrid orbitals)
- C- 2 and 3 show sp^2-hybridisaton (i.e. 3-hybrid orbital by each C-atom)
- C-4 show sp^3-hybridisation (i.e. 4-hybrid orbitals.
- N-atom show sp-hybridisation (i.e. 2-hybrid orbitals)

Thus, total number of hybrid orbitals

$= 2 + 6 + 4 + 2 = 14$

Hence, option (2) is correct.

Q.30 Answer:

1600 K

Solution:

$$P_1 = P_0 = 10^5\ \text{Pa} + \frac{1 \times 10}{10^{-4}} = 2 \times 10^5\ \text{Pa}$$

$$P_2 = P_0 + \frac{Mg}{A} = 10^5 + \frac{63 \times 10}{10^{-4}} = 64 \times 10^5\ \text{Pa}$$

For adiabatic process $\frac{T_1^\gamma}{P_1^{\gamma-1}} = \frac{T_2^\gamma}{P_2^{\gamma-1}}$

$$\Rightarrow \quad \frac{(400)^{\frac{5}{3}}}{(2\times10^5)^{\frac{2}{3}}} = \frac{T_2^{\frac{5}{3}}}{(64\times10^5)^{\frac{2}{3}}}$$

$$T_2 = 400(32)^{\frac{2}{5}} = 400 \times 2^2 = 1600\ \text{K}$$

$\therefore$ (D)

Q.31 Answer:

Maximum polarization is brought about by a cation of high charge

Solution:

According to Fajan's rule, polarisation of anion is influenced by charge and size of cation more is the charge and small size of cation causes more polarisation of anion having larger size and greater charge.

Q.32 Answer:

88.36 %

Solution:

For 750 mm pressure, subtract aqueous tension 30 mm to obtain pressure of nitrogen.

p = 750 − 30 = 720mm of Hg

Now, volume of nitrogen at STP $= \frac{V \times p \times 273}{T \times 760}$

$$= \frac{82 \times 720 \times 273}{300 \times 760}$$

$$= \frac{1.6117 \times 10^7}{228000} = \frac{1.6117 \times 10^4}{228}$$

$$= 70.68 \text{ mL}$$

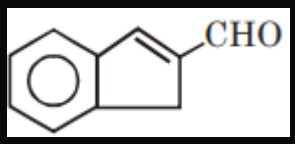
∴ Percentage of nitrogen

$$= \frac{\text{vol. of } N_2 \text{ at STP}}{\text{wt. of organic compound}} \times \frac{28}{22400} \times 100$$

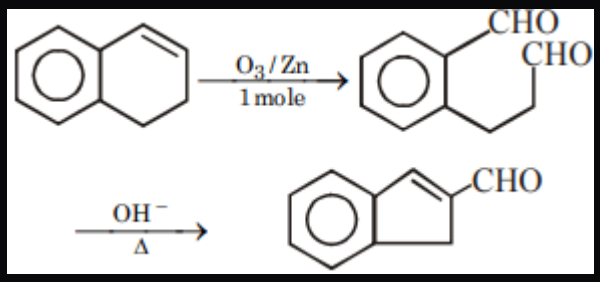
$$= \frac{70.68}{1} \times \frac{28}{22400} \times 100 = 88.36\%$$

The percentage composition of nitrogen in the compound would be 88.36%**.**

Q.33 Answer:



Solution:



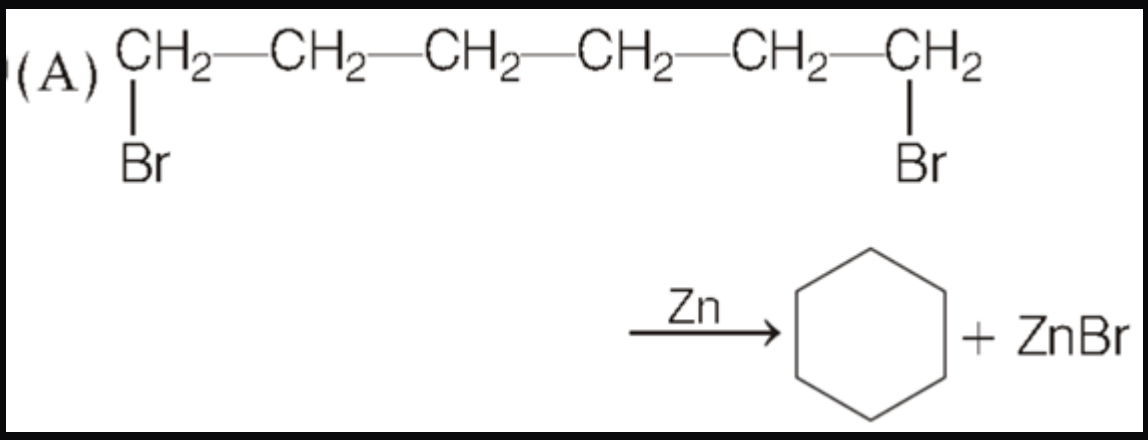
Intramolecular aldol

Q.34 Answer:

(a)

Solution:

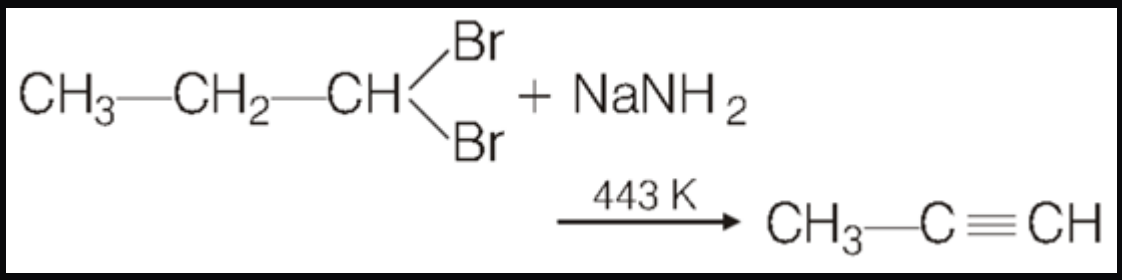
(a)



(B) $C_2H_5OH + H_2SO_4 \xrightarrow{443\ K} CH_2 = CH_2$

(C) $CH_3 - CH = CH_2 + HBr \xrightarrow[\text{Anti-Markownikof}]{\text{Peroxide}} CH_3 - CH_2 - CH_2 - Br$

(D)

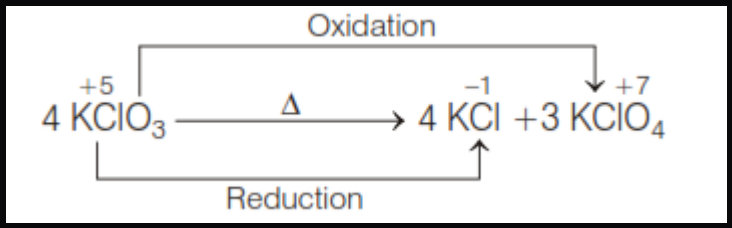


Q.35 Answer:

$4KClO_3(s) \xrightarrow{\Delta} KCl(s) + 3KClO_4(s)$

Solution:

Reaction (c) gives oxidation as well as reduction of same (one) species.



Q.36 Answer:

I, IV only

Solution:

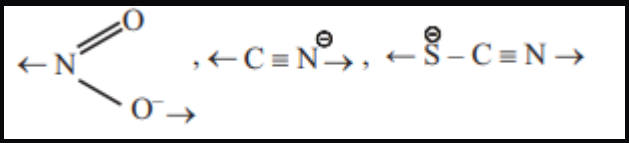
In the given reaction, Cl_2 is getting disproportionated into +1 state in ClO^- and -1 state in Cl^-.
Cl_2 to ClO^- is oxidation while Cl_2 to Cl^- is reduction.

Q.37 Answer:

$NO_2^{\ominus}, CN^{\ominus}, SCN^{\ominus}$

Solution:

The ligands containing two donor sites but use only one at a time are known as ambidentate ligand



Q.38 Answer:

(a) - (II), (b) - (II), (c) - (II)

Solution:

(a) $K_b \rightarrow II > I$

due to steric crowding at $-N$ atom

(b) $K_b \rightarrow II > I$

due to steric crowding at $-N$ atom

(c) $K_b \rightarrow II > I$

due to stability of conjugate acid

Q.39 Answer:

4

Solution:

(a) 'He' is the second most abundant element in the universe. The most abundant element in the universe is hydrogen, which makes up about three-quarters of all matter, Helium makes up most of the remaining 25%.

(b) Darmstadtium is a chemical element with the symbol Ds and atomic number 110. It is an extremely radioactive synthetic element.

(c) Under ordinary conditions of temperature and pressure, osmium is the element with the highest density. Its density is $22.59\ g/cm^3$.

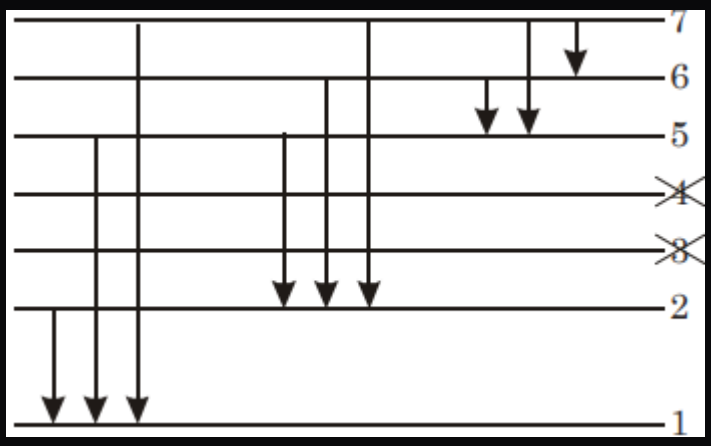
(d) Francium is the most electropositive element in the periodic table.

So, all these statements are correct.

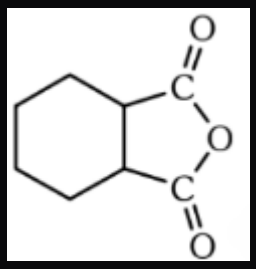
Q.40 Answer:

10

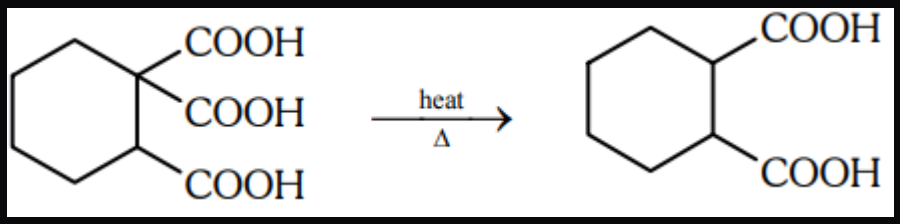
Solution:

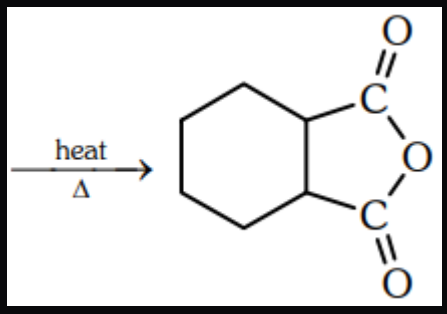


Q.41 Answer:

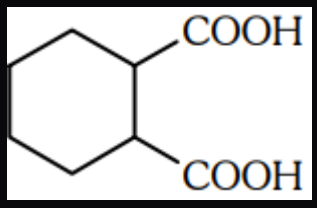


Solution:

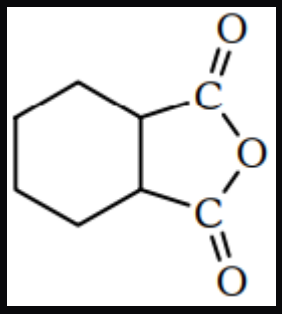




Geminal dicarboxylic acids undergo easy decarboxylation, hence 1st



is formed, then immediate dehydration takes place to form



Hence, choice (D) is correct while choices (A), (B) and (C) are incorrect.

Q.42 Answer:

paramagnetic $[Fe(H_2O)_5(NO)]SO_4$

Solution:

Cold FeSO, solution on absorption of NO develops brown colour due to the formation of $[Fe(H_2O)_5(NO)]SO_4$

$$FeSO_4 + NO + 5H_2O \rightarrow [Fe(H_2O)_5(NO)]SO_4$$

This complex has 3.89 BM magnetic moment shows that it has 3 unpaired electrons

Q.43 Answer:

5 mL

Solution:

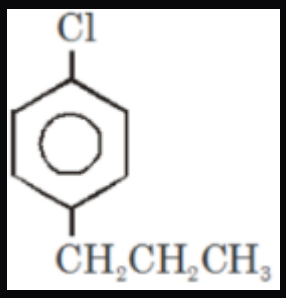
$$\text{gE of } H_2SO_4 = \text{gE of } AgNO_3 + \text{gE of NaOH}$$

$$\left(\frac{N_1\ V_1}{1000}\right)_{H_2SO_4} = \left(\frac{N_2\ V_2}{1000}\right)_{AgNO_3} + \left(\frac{N_3\ V_3}{1000}\right)_{NaOH}$$

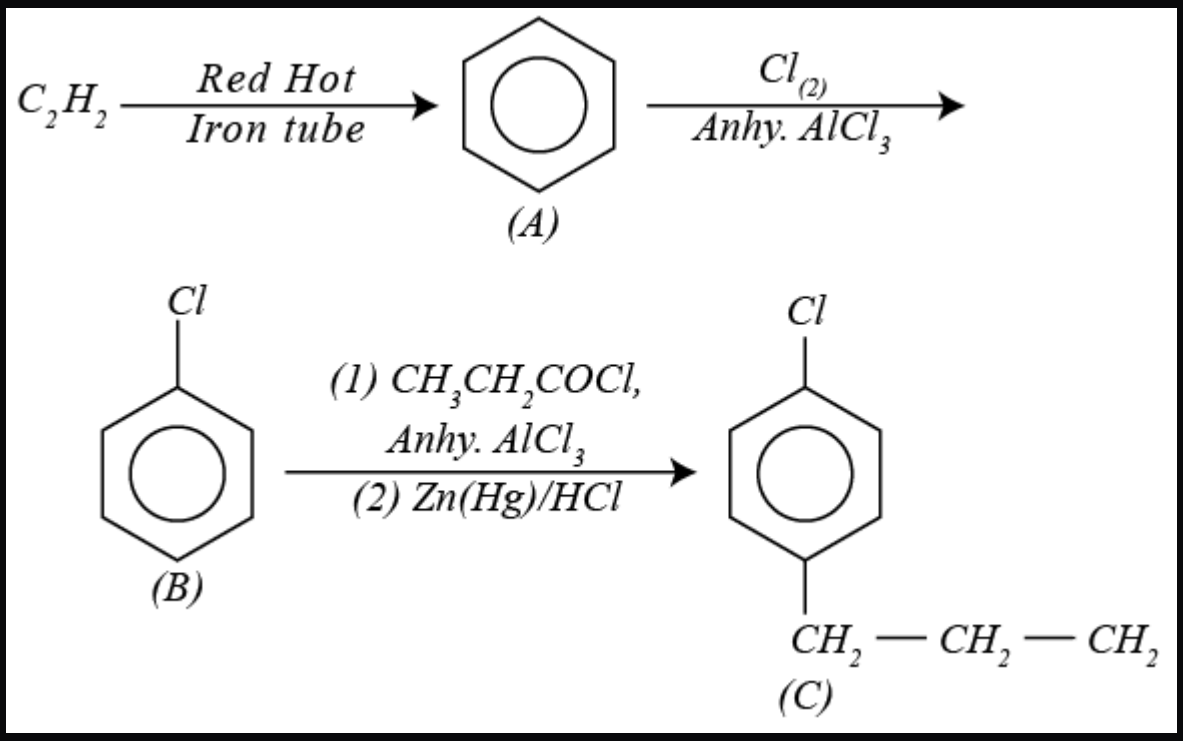
$$\left(\frac{\frac{2}{2} \times 20}{1000}\right)_{H_2SO_4} = \left(\frac{\frac{1}{2} \times 30}{1000}\right)_{AgNO_3} + \left(\frac{1 \times V_3}{1000}\right)_{NaOH}$$

$$V_3 = 5\ mL$$

Q.44 Answer:



Solution:



Q.45 Answer:

Fe^{2+} to Fe^{3+}

Solution:

This is usually done to convert Fe^{2+} to Fe^{3+} because $\mathrm{Fe}^{\mathrm{III}}$ hydroxides are insoluble which can be separated to a large extent frommore soluble than $\mathrm{Fe}^{\mathrm{II}}$ hydroxides.

Q.46 Answer:

—

Solution:

$$\begin{aligned} \% &= 62.07 & \frac{62.07}{12} &= 5.17 \rightarrow 3 \\ H &= 10.34 & \frac{10.34}{1} &= 10.34 \rightarrow 6 \\ O &= 27.59 & \frac{27.59}{16} &= 1.72 \rightarrow 1 \end{aligned}$$

$\therefore \quad C_3H_6\mathrm{O}$

$(\Delta U = 1)$

$\mathrm{Mol\ Wt} = 2 \times VD = 58$

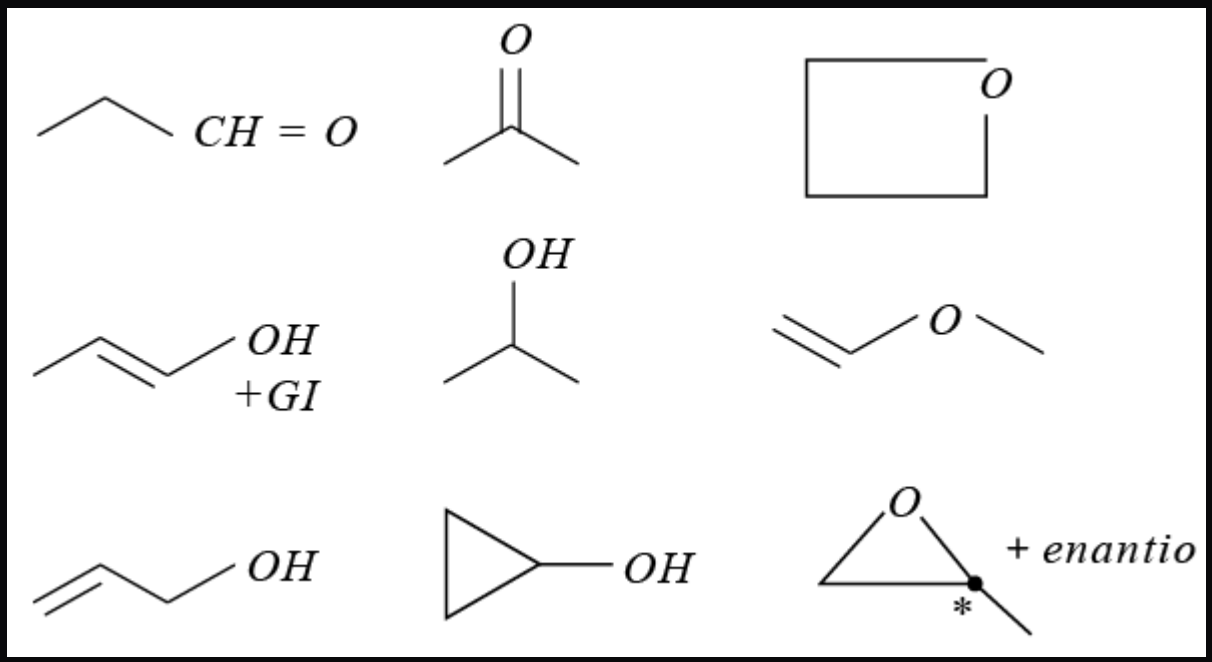
No $\mathrm{R_n}$ with Na metal $\longrightarrow \mathrm{NO-OH}$ group

Gives addition $R \times n$

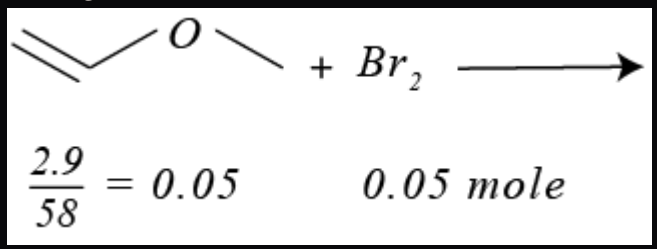
$\longrightarrow$ $c = c$ **is present**

$c = 0$ **is absent**

$\therefore$ its an unsaturated ether



$\therefore \quad y = 11$



$= \text{wt of } Br_2 \text{ used } = 0.05 \times 160$

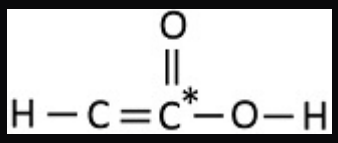
$= 8gm(\times)$

$\therefore \quad y - x = 11 - 8 = 3$

Q.47 Answer:

—

Solution:

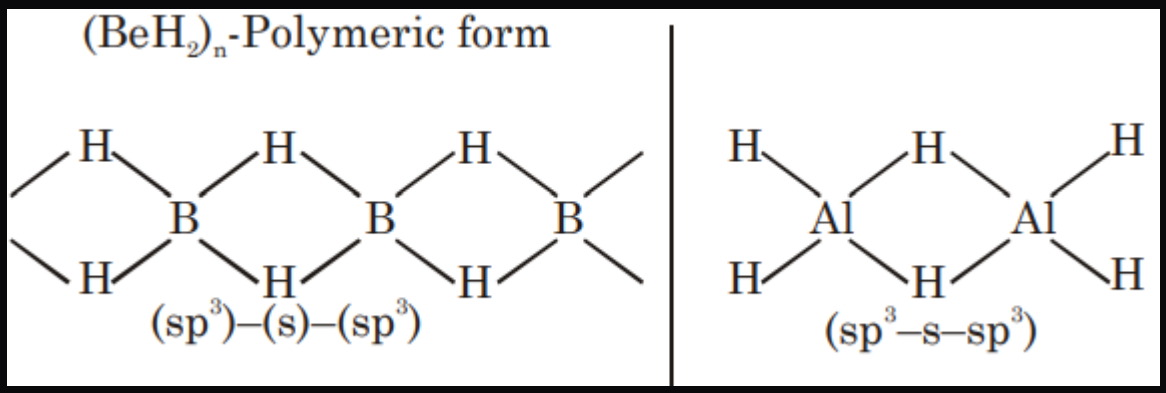


The star marked carbon has a valency of 5 and hence this formula is not correct.

Q.48 Answer:

—

Solution:



Q.49 Answer:

—

Solution:

$$\begin{array}{lccc} \mathrm{aA} & \longrightarrow & (\mathrm{b1})\mathrm{B} & + & \mathrm{C} \\ \mathrm{t}=0 & & & & \mathrm{a}' \\ & & & & \mathrm{a}' \propto \mathrm{P}_0 \\ \mathrm{t}=10\ \mathrm{min} & & \mathrm{a}'-\mathrm{x} & & \\ & & \frac{(b-1)}{a}x & & \end{array}$$

$$\frac{x}{a}\left[(a'-x)+\frac{(b-1)x}{a}+\frac{x}{a}\right] \propto P_0+P_0\frac{50\left(\frac{b}{a}-1\right)}{100}$$

$$\left[a'-x+\frac{b}{a}x\right] \propto P_0\left[1+\frac{1}{2}\left(\frac{b}{a}-1\right)\right]$$

$$\left[a'+x\left(\frac{b}{a}-1\right)\right] \propto P_0+\frac{P_0}{2}\left(\frac{b}{a}-1\right)$$

$$\Rightarrow x \propto \frac{P_0}{2}. \Rightarrow \quad x \propto \frac{a'}{2}.$$

$$\Rightarrow \text{ half life } = 10 \text{ minute.}$$

Q.50 Answer:

—

Solution:

Let the formula of hydrocarbon be $\mathrm{C_xH_y}$. Its combustion reaction is given as

$$\begin{array}{lcccc} & \mathrm{C_XH_y(g)} & +\left(\mathrm{x}+\frac{\mathrm{y}}{4}\right)\mathrm{O_2(g)} \rightarrow & \mathrm{xCO_2(g)}+ & \frac{\mathrm{y}}{2}\mathrm{H_2O(l)} \\ \text{Initial vol. in ml} & 5 & 30 & 0 & 0 \\ \text{Final volume} & 0 & 30-\left(\mathrm{x}+\frac{\mathrm{y}}{4}\right)5 & 5\mathrm{x} & 0 \end{array}$$

Out of $25\mathrm{ml}$ of resultant gas $10\mathrm{ml}$ was $\mathrm{CO_2}$ (absorbed by NaOH) and the remaining $15\mathrm{ml}$ was oxygen (absorbed by pyrogallol).

$$5x = 10; x = 2$$

$$30-\left(x+\frac{y}{4}\right)5 = 15$$

On solving, $\mathrm{y} = 4$

formula of gaseous hydrocarbon is $\mathrm{C_2H_4}$.

Q.51 Answer:

0

Solution:

$$7\left(xf'(x)+f(x)\right) = 3f(x)+4f(x+1)$$

$$7xf'(x) = 4(f(x+1)-f(x))$$

Let $f(x)$ be a degree n polynomial

LHS has degree n and RHS has degree

$$\mathrm{n}-1$$

$$\therefore \mathrm{f(x)} \text{ is constant}$$

$$\therefore \mathrm{f(x)} = 1$$

Q.52 Answer:

3

Solution:

$$S_1 > 0 \Rightarrow 2\cos^2\theta + 3\cos\theta - 2 > 0$$
$$(2\cos\theta - 1)(\cos\theta + 2) > 0$$
$$\cos\theta > \frac{1}{2}$$
$$\therefore \quad \theta \in \left[0, \frac{\pi}{3}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right]$$
$$\text{given } \theta = \frac{n\pi}{6} \Rightarrow \theta = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{12\pi}{6}$$
$$\therefore \quad n \in \{1, 11, 12\}$$

Q.53 Answer:

$(0, \infty)$

Solution:

$$f'(x) = \frac{(2 + 2^x - 3 \times 2^{-x})\ln 2}{(2^x + 1)^2} = \frac{(3 + 2^x)(1 - 2^{-x})}{(2^x + 1)^2}$$
$$\Rightarrow 1 - 2^{-x} > 0 \Rightarrow x > 0$$

Q.54 Answer:

$x^x = e^{ye^y}$

Solution:

$$\frac{dx}{dy}(1 + \log x) - x\log x = e^y$$
$$\text{Put } x\log x = t$$
$$\Rightarrow \frac{dt}{dy} - t = e^y$$
$$\text{Sol. is t.e } e^{-y} = y + C$$
$$\Rightarrow x\log x \cdot e^{-y} = y + C$$
$$y(1) = 0 \Rightarrow C = 0$$
$$\Rightarrow \log(x^x) = ye^y$$

Q.55 Answer:

det. $B = 12$

Solution:

Consider the det. B, using $R_1 \to R_1 + R_2 + R_3$

$$B = 2\begin{vmatrix} a+p+x & b+q+y & c+r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$$

using $R_2 \to R_2 - R_1$ and

$R_3 \to R_3 - R_1$

$$= 2\begin{vmatrix} a+p+x & b+q+y & c+r+z \\ -p & -q & -r \\ -x & -y & -z \end{vmatrix}$$

using $R_1 \to R_1 + R_2 + R_3$

$B = 2$ det. $A = 2 \cdot 6 = 12$

Q.56 Answer:

36

Solution:

Let $\vec{p} = a\hat{i} + b\hat{j} + c\hat{k}$

$\vec{q} = x\hat{i} + y\hat{j} + z\hat{k}$

Use Inequality

$$|\overrightarrow{\mathrm{p}} \times \overrightarrow{\mathrm{q}}| = |\overrightarrow{\mathrm{p}}||\overrightarrow{\mathrm{q}}| \sin\theta$$

$$|\overrightarrow{\mathrm{p}} \times \overrightarrow{\mathrm{q}}| \leqslant |\overrightarrow{\mathrm{p}}||\overrightarrow{\mathrm{q}}|$$

$$|\overrightarrow{\mathrm{p}} \times \overrightarrow{\mathrm{q}}|^2 = (\mathrm{az} - \mathrm{cx})^2 + (\mathrm{ay} - \mathrm{bx})^2 + (\mathrm{bz} - \mathrm{cy})^2$$

$$\leq \left(\mathrm{a}^2 + \mathrm{b}^2 + \mathrm{c}^2\right)\left(\mathrm{x}^2 + \mathrm{y}^2 + \mathrm{z}^2\right)$$

Q.57 Answer:

4

Solution:

As $\mathrm{Lim}_{x \to \pm\infty} f(x) = \infty$

$\therefore$ degree of $g(x)$ must be less than 4 .

As $\mathrm{Lim}_{x \to -1} f(x) = 6 \Rightarrow \mathrm{Lim}_{x \to -1} \frac{\left(x^2-1\right)\left(x^2-4\right)}{g(x)} = 6 \Rightarrow g(x)$ must have $(x+1)$ as a factor.

again, $\mathrm{Lim}_{x \to -2} f(x) \neq 0$ but finite $\Rightarrow g(x)$ must have $(x+2)$ another factor.

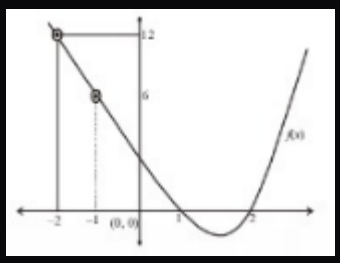
$\therefore g(x) = \lambda(x+1)(x+2)$

Now, $\mathrm{Lim}_{x \to -1} \frac{\left(x^2-1\right)\left(x^2-4\right)}{\lambda(x+1)(x+2)} = 6$

$\Rightarrow \mathrm{Lim}_{x \to -1} \frac{(x-1)(x-2)}{\lambda} = 6 \Rightarrow \frac{(-2)(-3)}{\lambda} = 6 \Rightarrow \lambda = 1$

Graph of $f(x)$:

$$f(x)=\frac{\left(x^2-1\right)\left(x^2-4\right)}{(x+1)(x+2)} x \in R-\{-2,-1\}$$
$$=(x-1)(x-2) x \in R-\{-2,-1\}$$



Q.58 Answer:

16

Solution:

Since a, b, c, d are in A.P.

$\therefore b-a=c-b=d-c=D$ (let common difference)

$\Rightarrow d=a+3D$

$\Rightarrow a-d=-3D$ and $d=b+2D \Rightarrow b-d=-2d$

Also, $c=a+2D \Rightarrow c-a=2D$

$\therefore$ Given equation $2(a-b)+k(b-c)^2+(c-a)^3=2(a-d)+(b-d)^2+(c-d)^3$

Becomes $-2D+kD^2+(2D)^3=-6D+4D^2-D^3$

$\Rightarrow 9D^2+(k-4)D+4=0$

Since D is real $\Rightarrow (k-4)^2-4(4)(9) \geq 0$

$\Rightarrow \mathrm{k}^2-8\mathrm{k}-128 \geq 0 \Rightarrow (\mathrm{k}-16)(\mathrm{k}+8) \geq 0$

$\therefore \mathrm{k} \in (-\infty,-8] \cup [16,\infty)$

Hence smallest positive value of $k=16$

Q.59 Answer:

$a=1.25,\ b=-5$

Solution:

$w_i=ax_i+b$

$\Rightarrow \bar{w}=a\bar{x}+b$

$\Rightarrow 60=a(52)+b \ldots$

S. D of $w=|a|$ (**S. D of** x_i)

$15=|a|12$

$\Rightarrow a=\pm 1.25 \ldots$

From (i) and (ii), we get,

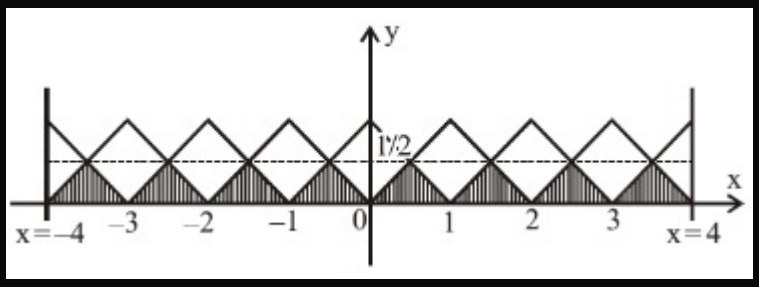
If $a=1.25 \Rightarrow 60=65+b \Rightarrow b=-5$

If $a=-1.25 \Rightarrow 60=-65+b \Rightarrow b=125$

Q.60 Answer:

2

Solution:



Shaded area = required area $= 8 \times \frac{1}{2} \times \frac{1}{2} = 2$ **sq. units**

Q.61 Answer:

50

Solution:

Given, $f(y) = f(x) + y$

$\Rightarrow y^2 + ay + b = x^2 + ax + b + y$

$\Rightarrow y^2 + y(a-1) - x^2 - ax = 0$

As, $y \in R$**, so** $D \geq 0 \quad \forall x \in R$

$\Rightarrow (a-1)^2 + 4\left(x^2 + ax\right) \geq 0 \quad \forall x \in R$

$\Rightarrow 4x^2 + 4ax + a^2 - 2a + 1 \geq 0 \quad \forall x \in R$

Now, $\mathrm{D} \leq 0$

$\Rightarrow 16a^2 - 16\left(a^2 - 2a + 1\right) \leq 0$

$\Rightarrow 2a - 1 \leq 0 \Rightarrow a \leq \dfrac{1}{2}$

So, $\mathrm{a}_{\max.} = \frac{1}{2}$

Hence, maximum value of $100a = 50$

Q.62 Answer:

$\frac{2}{\sqrt{5}}$ units

Solution:

$\ell_1 : \bar{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$

$= (\hat{i} - 2\hat{j} + 3\hat{k}) + t(-\hat{i} + \hat{j} - 2\hat{k})$

$\ell_2 : \bar{r} = (\hat{i} - \hat{j} + \hat{k}) + p(\hat{i} + 2\hat{j} + 2\hat{k})$

Here $\bar{a}_2 - \bar{a}_1 = (\hat{i} - \hat{j} + \hat{k}) - (\hat{i} - 2\hat{j} + 3\hat{k}) = \hat{j} - 2\hat{k}$

$$\bar{b}_1 \times \bar{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 1 & 2 & 2 \end{vmatrix} = \hat{i}(2+4) - \hat{j}(0) - 3\hat{k} = 6\hat{i} - 3\hat{k}$$

$\therefore \left|\bar{b}_1 \times \bar{b}_2\right| = \sqrt{36+9} = 3\sqrt{5}$

$$\text{shortest distance } = \left|\frac{(\bar{b}_1 \times \bar{b}_2) \cdot (\bar{a}_2 - \bar{a}_1)}{(\bar{b}_1 \times \bar{b}_2)}\right| = \left|\frac{(6\hat{i} - 3\hat{k}) \cdot (\hat{j} - 2\hat{k})}{3\sqrt{5}}\right| = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}}$$

Q.63 Answer:

$\arg\left(\frac{z+i}{z-3}\right) = \frac{\pi}{6}$

Solution:

The image of z in the real axis is $\bar{z}$.

$\therefore$ The image is given by $\arg\left(\frac{\bar{z}-3}{\bar{z}-i}\right) = \frac{\pi}{6}$. But $\arg \bar{z} = -\arg z$

$\therefore \quad \arg\left(\frac{\bar{z}-3}{\bar{z}-i}\right) = \frac{\pi}{6} \Rightarrow \arg\left(\frac{z-3}{z+i}\right) = -\frac{\pi}{6} \arg\left(\frac{z+i}{z-3}\right) = \frac{\pi}{6}$

Q.64 Answer:

$\frac{5\pi}{3}$

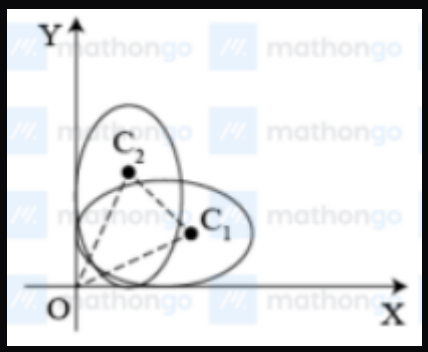
Solution:

Clearly, origin will be a point lying on the director circle of the ellipse.

Thus, equation of the locus of centre of ellipse will be

$$x^2 + y^2 = a^2 + b^2$$

$$\Rightarrow \quad x^2 + y^2 = 100$$



The adjacent figure shows the position of centre of ellipse in two extreme position

$Q = \angle C_1OC_2$

$= \frac{\pi}{2} - 2\tan^{-1}\left(\frac{5}{5\sqrt{3}}\right)$

$= \frac{\pi}{6}$

Thus, are length $C_1C_2 = 10 \cdot \frac{\pi}{6} = \frac{5\pi}{3}$

Q.65 Answer:

3

Solution:

$(A-B)^3 = A^3 - B^3 - 3A^2B + 3A^2$

$= A - B - 3AB + 3AB$

$(A-B)^3 = (A-B)$

$|A-B|^3 = |A-B|$

$\Rightarrow |A-B| = 0, 1 \text{ or } -1$

Q.66 Answer:

1405

Solution:

Case-I: All six digits alike $\times \times \times \times \times \times$.

say 111111,22222 etc $= 5$ **ways**

Case-II: 2 alike +2 other alike +2 other alike.

$^5\mathrm{C}_3$ **say** $1, 2, 3$ **and take** $11, 22, 33$

$\therefore$ **Number of ways** $= \frac{6!}{2!2!2!}$ **(in each selection)**

$\Rightarrow \text{Total } = 10\left(\frac{720}{8}\right) = 900$

Case-III: 2 alike +4 other alike e.g. 112222 or 221111 etc.

Number of ways selecting 2 digits $= \left(^5\mathrm{C}_2\right)(2)$ **(in each selection)**

$\therefore$ **Number of ways** $= \frac{6!}{2!4!}$

$\Rightarrow \text{Total } = \left(^5\mathrm{C}_2\right)(2)\left(\frac{6!}{2!4!}\right) = (20)(15) = 300$

Case-IV: 3 alike +3 other alike.

Select $^5C_2$ **e.g. 111,222 etc.**

Total ways $^5C_2 \cdot \frac{6!}{3!\cdot 3!} = 200$

$\text{Total } = 5 + 900 + 300 + 200 = 1405$

Q.67 Answer:

$\frac{1}{33}$

Solution:

T **: took a taxi**

R **: Took a rickshaw**

W : Walked by foot

$\mathrm{P(T)}=\frac{1}{2},\mathrm{P(R)}=\frac{1}{3},\mathrm{P(W)}=\frac{1}{6}$

C : Catching the train

$$\mathrm{P(C/R)}=\frac{1}{2}\mathrm{P(C/T)};\mathrm{P(C/W)}=\frac{1}{4}\mathrm{P(C/R)}$$

$$\text{let } \mathrm{P(C/R)}=\mathrm{x}$$

$$\therefore \mathrm{P(C/T)}=2\mathrm{x};\mathrm{P(C/W)}=\frac{\mathrm{x}}{4}$$

$$\mathrm{P(W/C)}=\frac{\frac{1}{6}\times\frac{\mathrm{x}}{4}}{\frac{1}{2}\times 2\mathrm{x}+\frac{1}{3}\times\mathrm{x}+\frac{1}{6}\times\frac{\mathrm{x}}{4}}$$

$$=\frac{\frac{1}{24}}{1+\frac{1}{3}+\frac{1}{24}}=\frac{1}{24+8+1}=\frac{1}{33}$$

Q.68 Answer:

4

Solution:

$$\tan^4 x+\cot^4 x+1=3\sin^2 y$$

$$\Rightarrow \left(\tan^2 x-\cot^2 x\right)^2+3=3\sin^2 y$$

$\because$ L.H.S ≥ 3 and R.H.S ≤ 3 so solution exist only if

L.H.S. = R.H.S $=3\Rightarrow \tan^2 x=\cot^2 x$ **and** $\sin 2\mathrm{y}=1$

$$\Rightarrow \quad \mathrm{x}=\pm\frac{\pi}{4},\pm\frac{3\pi}{4}\ldots\ldots\&\mathrm{y}=\pm\frac{\pi}{2},\pm\frac{3\pi}{2},\ldots\ldots$$

but out of above solutions only $(x,y)=\left(\pm\frac{\pi}{4},\pm\frac{\pi}{2}\right)$ **lies inside the circle, therefore four pair of solutions exist.**

Q.69 Answer:

$\frac{1}{2\sqrt{2}}$

Solution:

We have, the curve $y^2-x^2=1$ **It abscissa** $=n$ **and ordinate** $=\sqrt{1+n^2}$

$\therefore P$ **is** $\left(n,\sqrt{1+n^2}\right)$

$d_n=$ **Shortest distance = Perpendicular distance of the line** $y=x$

$$\Rightarrow dx=\frac{\sqrt{1+n^2}-n}{\sqrt{2}}=\lim_{n\to\infty}\frac{n\left(\sqrt{1+n^2}-n\right)}{\sqrt{2}}\times\frac{\sqrt{1+n^2}+n}{\sqrt{1+n^2}+n}$$

$$\therefore \lim_{n\to\infty}\frac{n}{\sqrt{2}\left[\sqrt{1+n^2}+n\right]}=\frac{1}{2\sqrt{2}}$$

Q.70 Answer:

$\frac{1}{6}(1+3\sqrt{5})$

Solution:

$$\int_{-1}^{1}\{x^2+x-3\}\,dx=\int_{-1}^{1}\{x^2+x\}\,dx$$
$$=\int_{-1}^{1}\left(x^2+x-\left[x^2+x\right]\right)dx$$
$$=\left[\frac{x^3}{3}+\frac{x^2}{2}\right]_{-1}^{1}-\int_{-1}^{0}\left[x^2+x\right]dx-\int_{0}^{\frac{\sqrt{5}-1}{2}}\left[x^2+x\right]dx-\int_{(\sqrt{5}-1)y_2}^{1}\left[x^2+x\right]dx$$
$$=\frac{2}{3}+1-0-1\left(1-\frac{\sqrt{5}-1}{2}\right)\left[x^2+x\right]dx$$
$$=\frac{2}{3}+\frac{\sqrt{5}-1}{2}$$
$$=\frac{1}{6}(1+3\sqrt{5})$$

Q.71 Answer:

—

Solution:

$$x\,\mathrm{d}y=y(\,\mathrm{d}x+y\,\mathrm{d}y)$$
$$\Rightarrow y\,\mathrm{d}x=\left(x-y^2\right)\mathrm{d}y\Rightarrow\frac{\mathrm{d}x}{\mathrm{d}y}+\left(-\frac{1}{y}\right)x=-y$$
$$\therefore\quad \text{I.F.}=\mathrm{e}^{\int-\frac{1}{dy}y^y}=\mathrm{e}^{-\log y}=\frac{1}{y}$$

$\therefore$ **Solution of the given equation is**

$$x\cdot\frac{1}{y}=\int-y\cdot\frac{1}{y}\,\mathrm{d}y+\mathrm{c}$$
$$\Rightarrow\frac{x}{y}=-y+\mathrm{c}\ldots.(\mathrm{i})$$

Since $y(1)=1$, i.e., $y=1$ when $x=1$

$$\therefore\quad 1=-1+c\Rightarrow c=2$$
$$\therefore\quad \frac{x}{y}=-y+2\quad\ldots[\text{From (i)}]$$

Putting $x=-3$, we get

$$-\frac{3}{y}=-y+2$$
$$\Rightarrow y^2-2y-3=0$$
$$\Rightarrow(y-3)(y+1)=0$$

Since $y(x)>0, y=3$

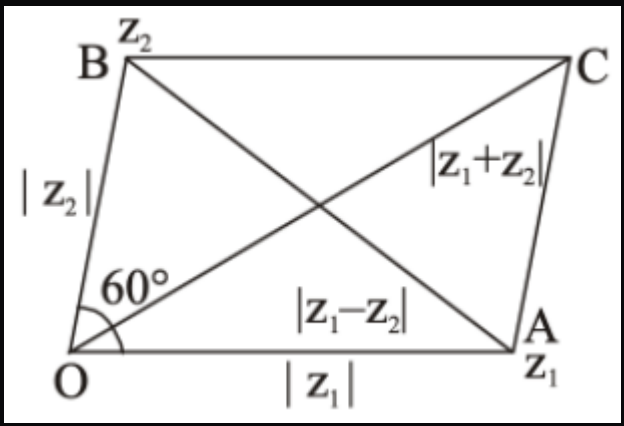
Q.72 Answer:

—

Solution:

Using cosine rule

$$|z_1+z_2| = \sqrt{|z_1|^2+|z_2|^2-2|z_1||z_2|\cos 120^\circ}$$
$$= \sqrt{4+9+2\cdot 3} = \sqrt{19}$$
$$\text{and } |z_1-z_2| = \sqrt{|z_1|^2+|z_2|^2-2|z_1||z_1|\cos 60^\circ}$$
$$= \sqrt{4+9-6} = \sqrt{7}$$
$$\therefore \left|\frac{z_1+z_2}{z_1+z_2}\right| = \sqrt{\frac{19}{7}} = \frac{\sqrt{133}}{7} \Rightarrow N = 133$$



Q.73 Answer:

—

Solution:

$$\text{Let } P(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + \ldots. + a_1$$
$$P(2x) = a_0 2^n x^n + a_1 2^{n-1}x^{n-1} + \ldots. + a_n$$
$$P(x) + P(2x) = 5x^2 - 18$$
$$\Rightarrow (a_0 + a_0 2^n)\, x^n + \ldots + 2a_n = 5x^2 - 18$$
$$\therefore n = 2$$
$$(a_0 + 4a_0)\, x^2 + 2a_2 = 5x^2 - 18$$
$$5a_0 = 5 \text{ and } a_2 = -9$$
$$\therefore P(x) = x^2 - 9$$
$$\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3}(x+3) = 6$$

Q.74 Answer:

—

Solution:

The given line passes through $A(5,-2,6)$

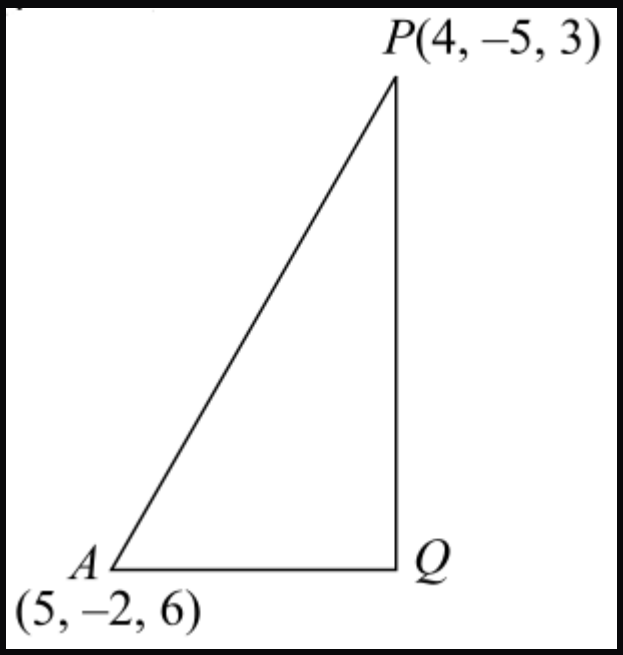
$$(PQ)^2 = (AP)^2 - (AQ)^2$$
$$(AP)^2 = (4-5)^2 + (-5+2)^2 + (3-6)^2 = 19$$

AQ **is the projection of** AP **on the given line**

$$\therefore AQ = (5-4)\frac{3}{\sqrt{50}} + (-2+5)\frac{(-4)}{\sqrt{50}} + (6-3)\left(\frac{-5}{\sqrt{50}}\right) = \frac{6}{\sqrt{50}}$$

$$\Rightarrow (PQ)^2 = 19 - \frac{36}{50} = \frac{457}{25}$$

$$\Rightarrow 25(PQ)^2 - 450 = 7$$



Q.75 Answer:

—

Solution:

$N_1 + N_2 + N_3 + N_4 + N_5 =$ **Total ways** $-\{$**when no odd**$\}$

Total ways $= {}^9C_5$

Number of ways when no odd number is selected is $0(\because$ **only available even are** $2, 4, 6, 8)$

$\therefore {}^9C_5 - 0 = 126$